MME 351 - Marine Electrical Technology

(2 0 2)

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Course aim

- To provide insight into electrical technology used in marine vessels.







Course Content

- **Power generation and distribution:** Selection of DC and AC generators, Merits and Demerits of AC and DC. Construction & Characteristics of brushless high speed AC generators with automatic Voltage regulator.
- **Rules and Regulations governing electrical machinery on ships**, location and installation of generator sets. Rating and characteristics of main switch boards. Arrangements of preferential trips and protective devices with wiring diagrams.
- **System earthing:** Isolated neutral systems, resistance earthed system.

Course Content

- **Motor and Control Equipment:** Special requirements of motors and starters for anchor Windlass & Capstan, wiring diagram, lowering & hoisting arrangement, overload protection, magnetic disc, brakes. Centralised control of motors in machinery spaces. Classes of insulation (A, B and E) and Temperature rise.
- **Starters for miscellaneous electrical equipment** for machinery space auxiliaries eg. Sequential starting and cut outs for an automatic fired boiler incorporating safety devices & combustion control equipment. Sequential starting for refrigerating plants, incorporating safety devices like high pressure cut out, cooling water failure cut out, automatic defrosting with help of timer.

Assessment

- **Continuous assessment** (Assignments, quizzes, mid-sem) - 30%
- **End-of-Semester Examination** (MCQs, Written paper) - 70%

Power Generation and Distribution in Marine Vessels



Uses of electricity in watercraft



Lighting



Pumps

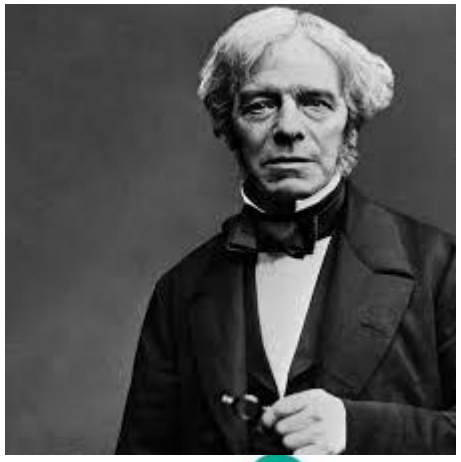


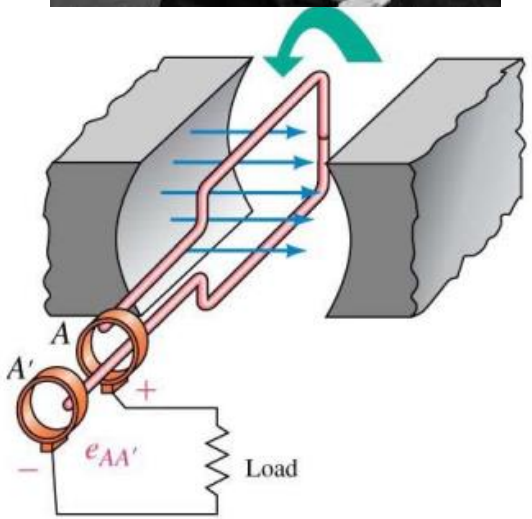


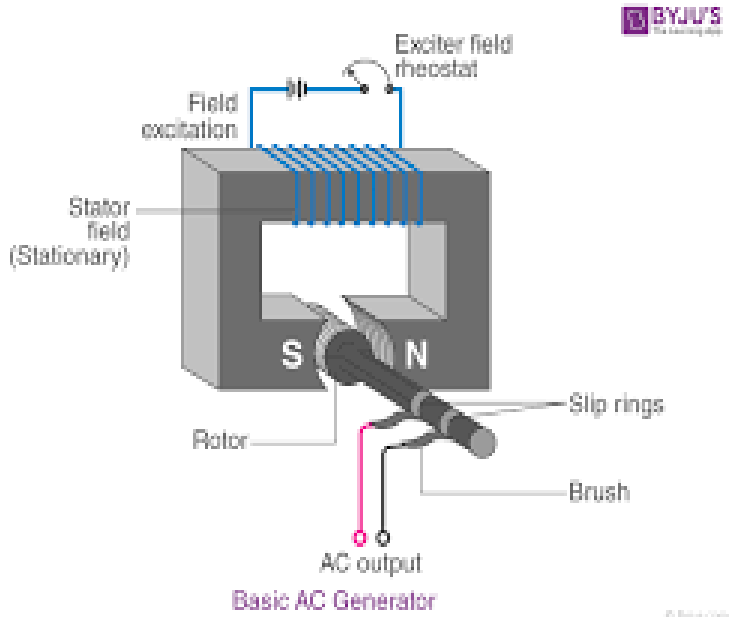
Refrigeration

AC generators

Operating principle of AC generators





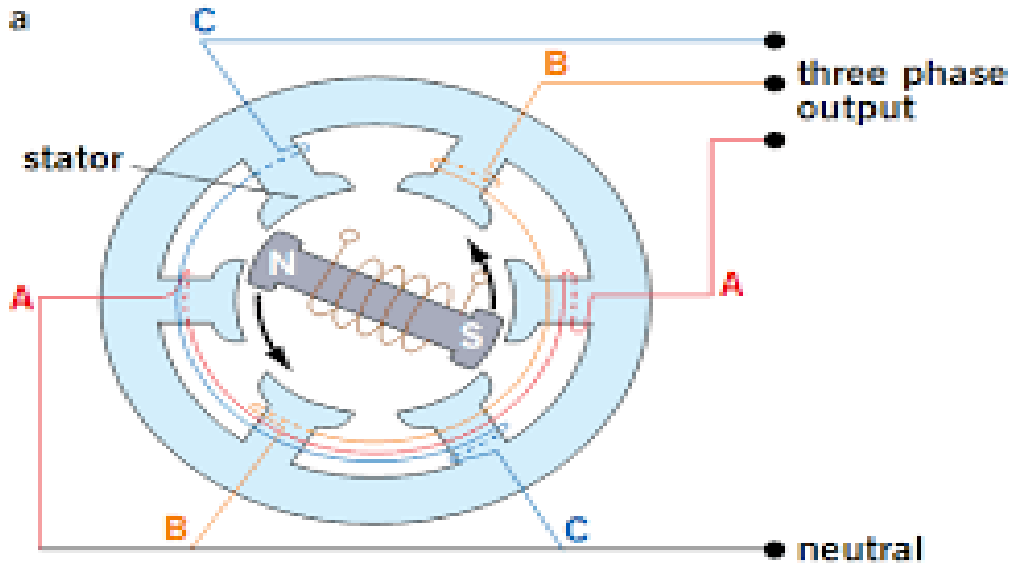




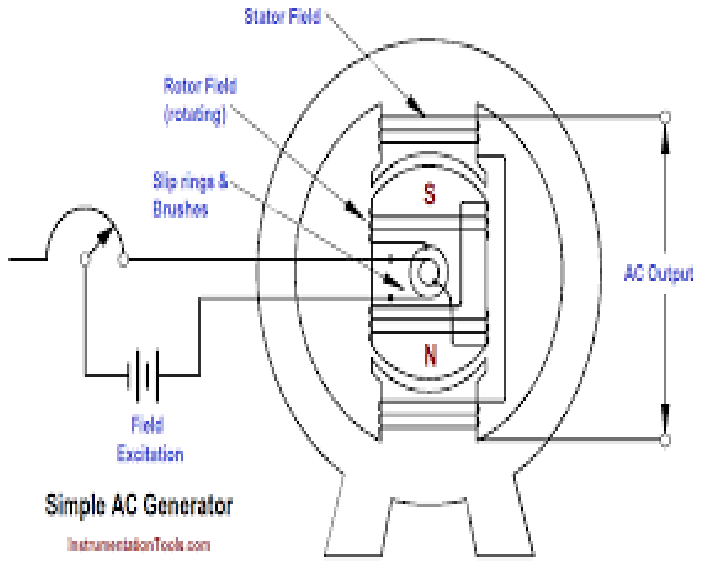


AC generators

Types AC generators



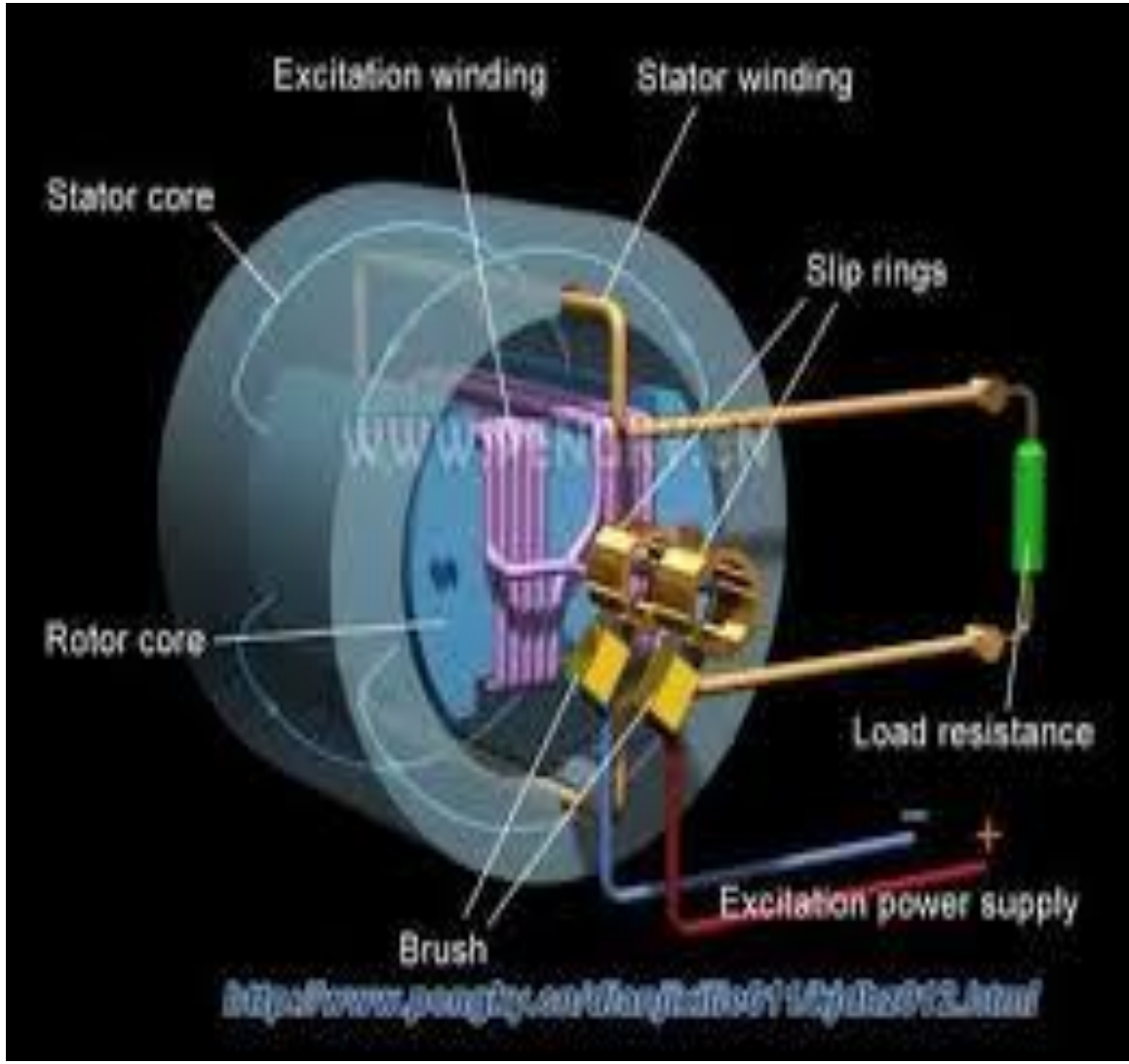
3-phase AC generator



1-phase AC generator

AC generators

- **Major components**
  - **Stator**
  - **Rotor**
  - **Slip rings**

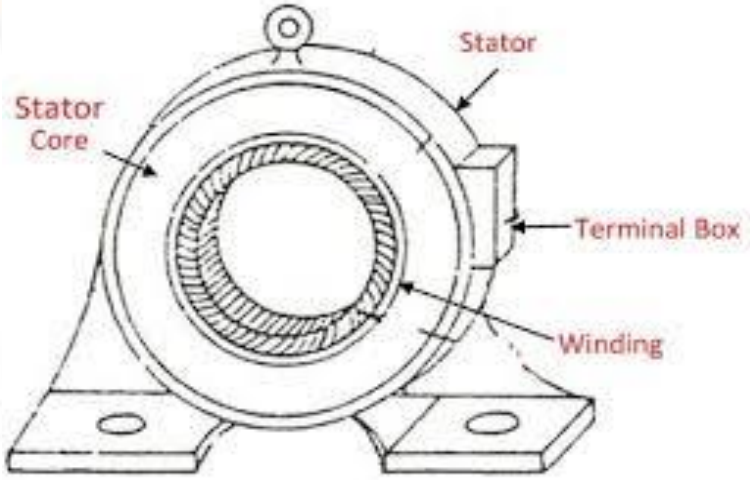


AC generators

- **Stator**











AC generators

- **Rotor**

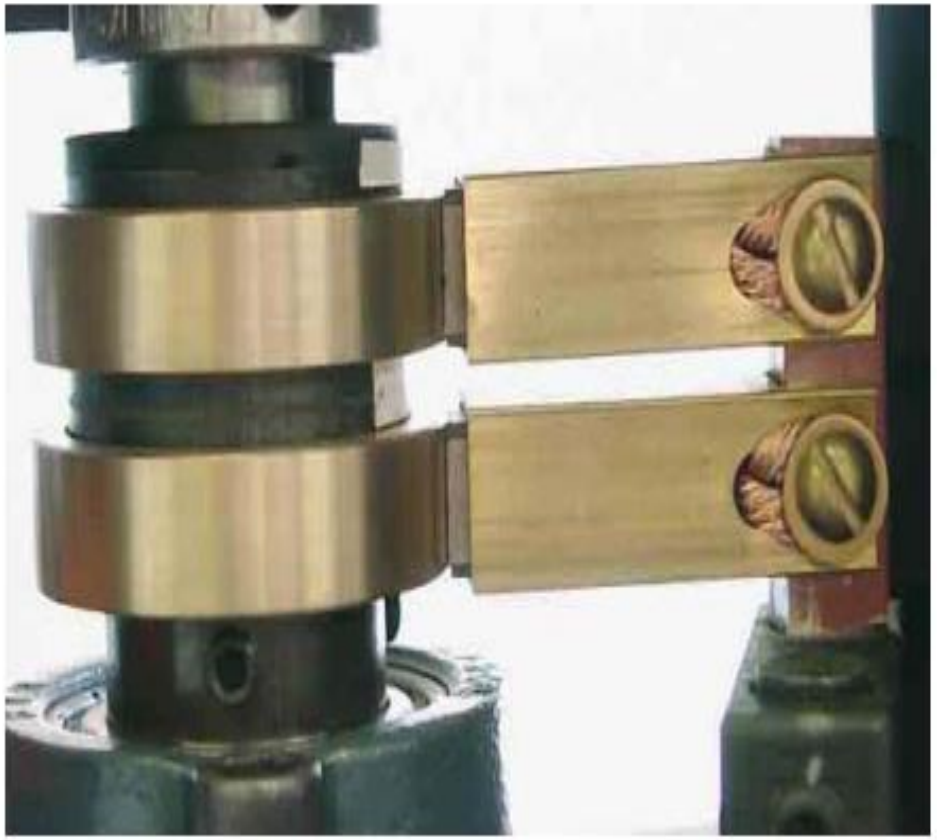






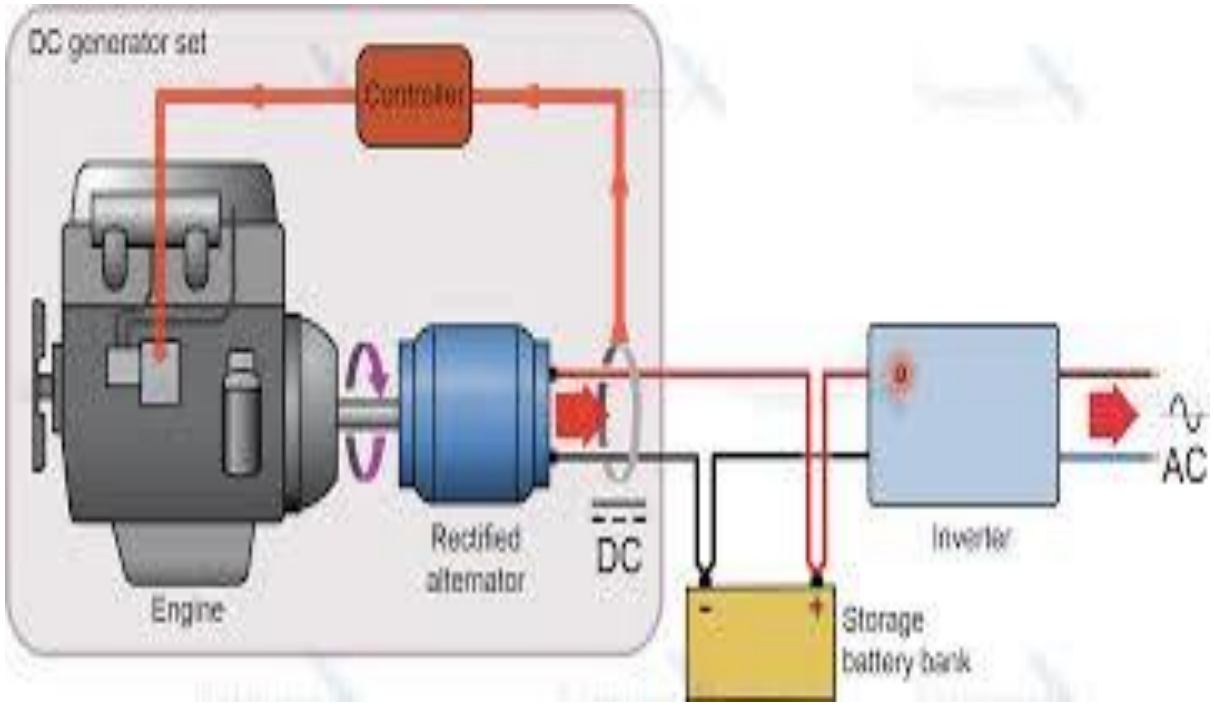
AC generators

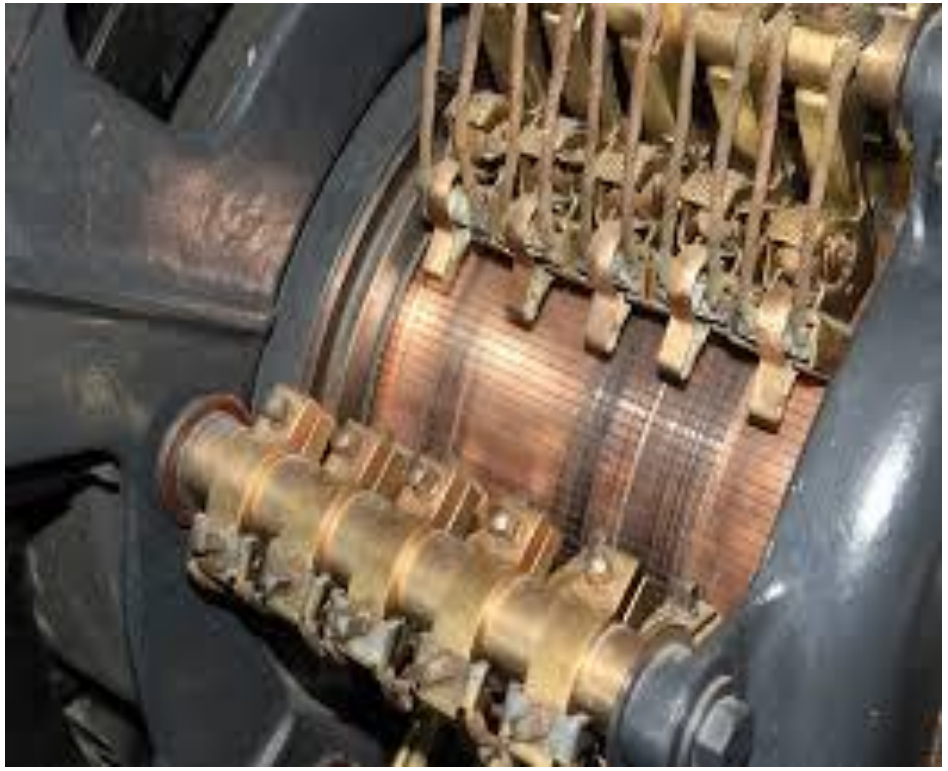
- **Slip rings**



DC generators

- **A DC generator is essentially a 1-phase AC generator with commutator**





AC vrs DC generators

- **DC Generators**
  - Suitability for large motors
  - Simple design
- **AC Generators**
  - Suitability for electrical appliances and small motors
  - Low maintenance needs
  - Easy distribution of the output with a transformer
  - Higher efficiency
  - Losses are relatively fewer than DC Generators
  - AC generators can be easily stepped up and stepped down through transformers.
  - Smaller in sizer compared to DC generators.

Emergency power supply

- Emergency power is needed when the main supply system fails.
- Emergency power can be supplied by
  - Batteries
  - Emergency generator
  - Both batteries and generator
- Emergency power should cover the following key components and systems
  - Steering gear system
  - Emergency bilge pump
  - Watertight doors
  - Fire fighting system
  - Ship's navigation lights and emergency lights
  - Communication and alarm system



Location and installation of generators

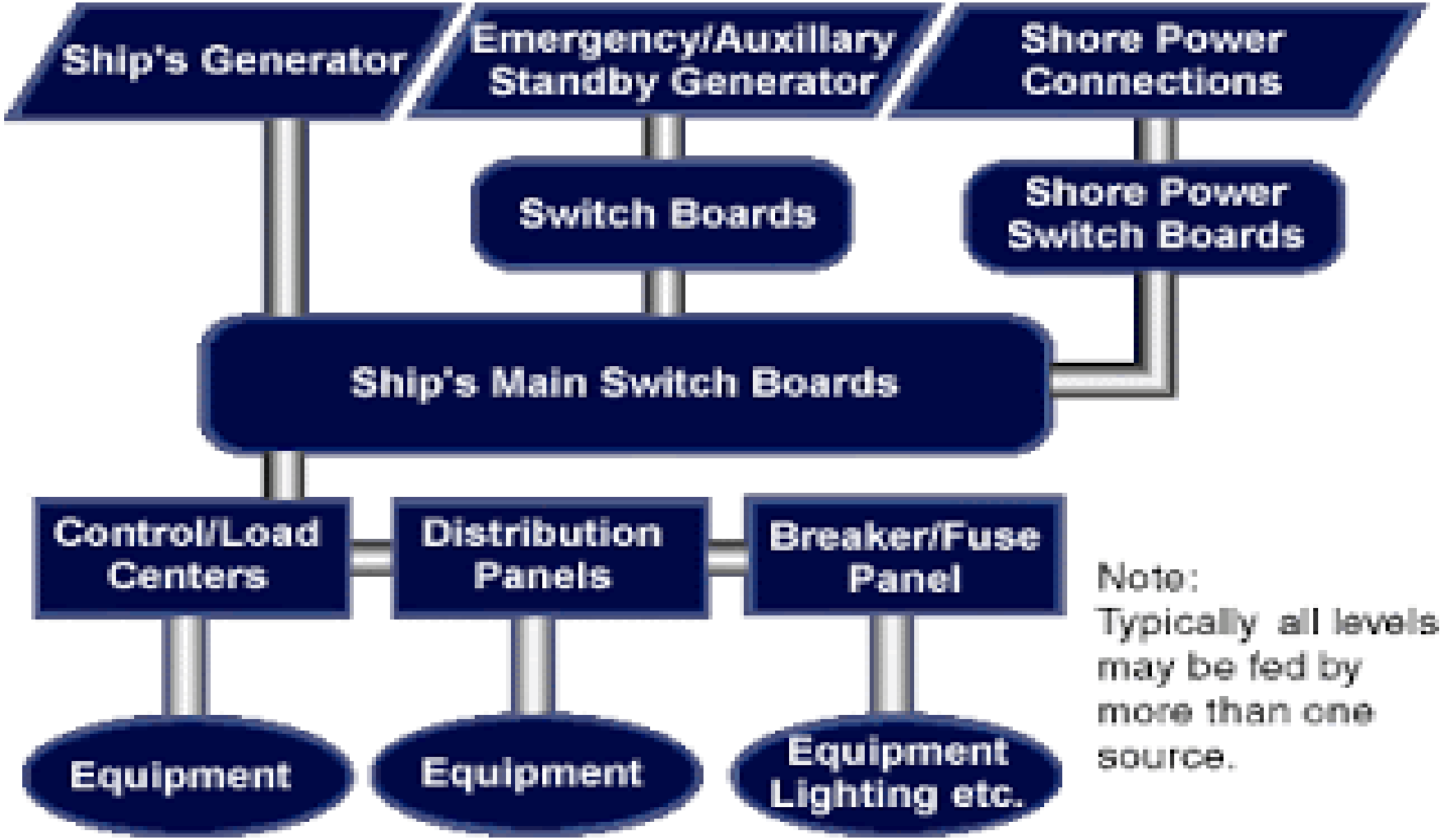
- A solid, level mounting platform is necessary
- Adequate space on all sides for ventilation and servicing
- Generators must be kept away from living quarters, and away from bilge splash and vapors.

Emergency power supply

- **Emergency generator is normally located outside the machinery space to provided unhindered access to it.**

Electrical distribution systems

- The power distribution system of a ship is designed to provide a secure supply to all loads with built-in protection for equipment and personnel.



Electrical distribution systems

A ship's electricity distribution system consists of different component for distribution and safe operation of the system. They are:

- Ship generator consisting of prime mover and alternator.
- Main switch board which is a metal enclosure taking power from a generator and supplying it to different machinery.
- Bus Bars which act as a carrier and allow transfer for transfer of power from one point to another.
- Circuit breakers which act as a switch and in unsafe condition can be tripped to avoid breakdown and accidents.
- Fuses as safety device for machinery.
- Transformers to step up or step down the voltage.

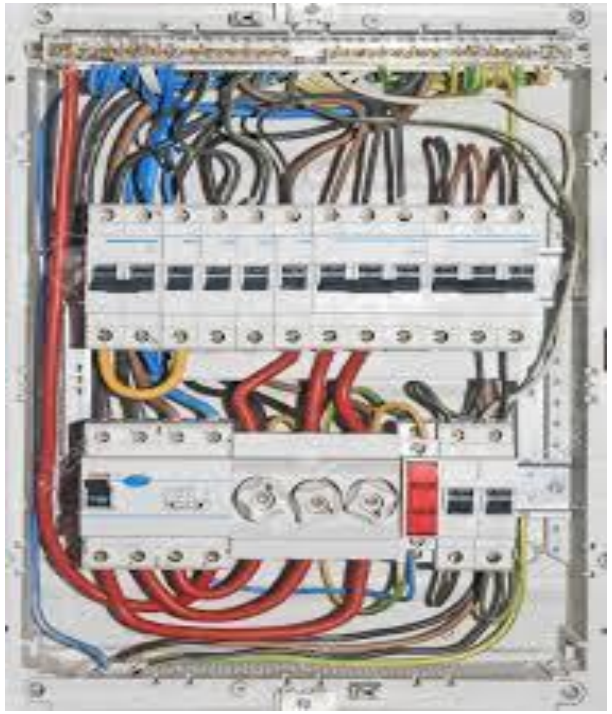
Electrical distribution systems

- **Generator**



Electrical distribution systems

- **Main switch board**



Domestic



Ship

Electrical distribution systems

- **Busbars**







Electrical distribution systems

- **Fuse**



Electrical distribution systems

- **Circuit breaker**







Electrical distribution systems

- **Transformer**



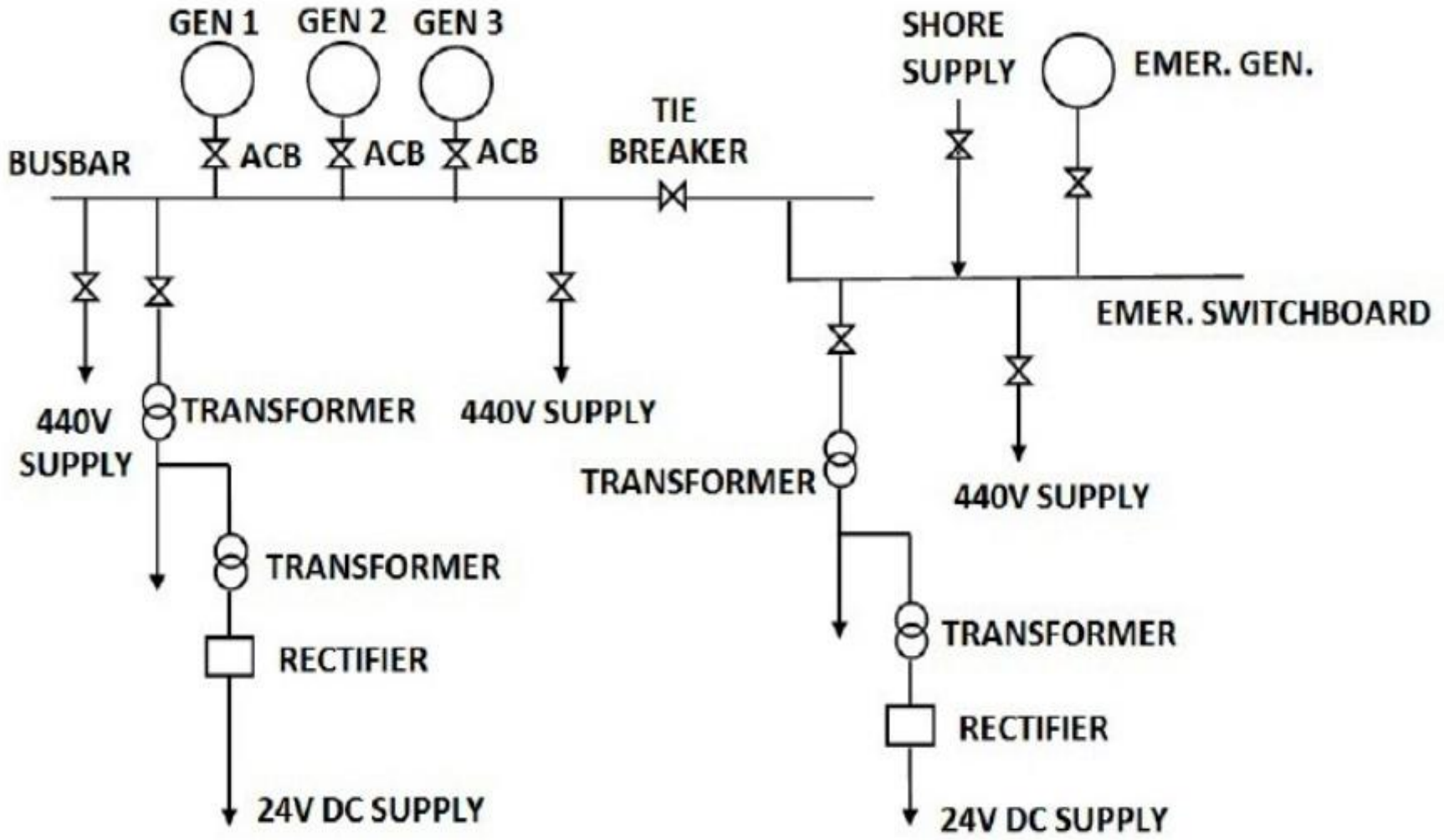








Electrical distribution systems



A typical distribution system for a ship

Emergency switchboard

- It is the emergency power distribution center.
- It distributes power to auxiliaries that are vital, under emergency conditions
- Under normal operation, power for the emergency switchboard is supplied by the main generators through a bus-tie.
- In case of power failure, loss of voltage at the emergency switchboard will cause the emergency generator to automatically start.
- Simultaneously, the bus-tie will open disconnecting the main and the emergency switchboards.
- The emergency generator breaker will close, energizing the emergency board and providing the needed emergency power.

Electrical distribution systems

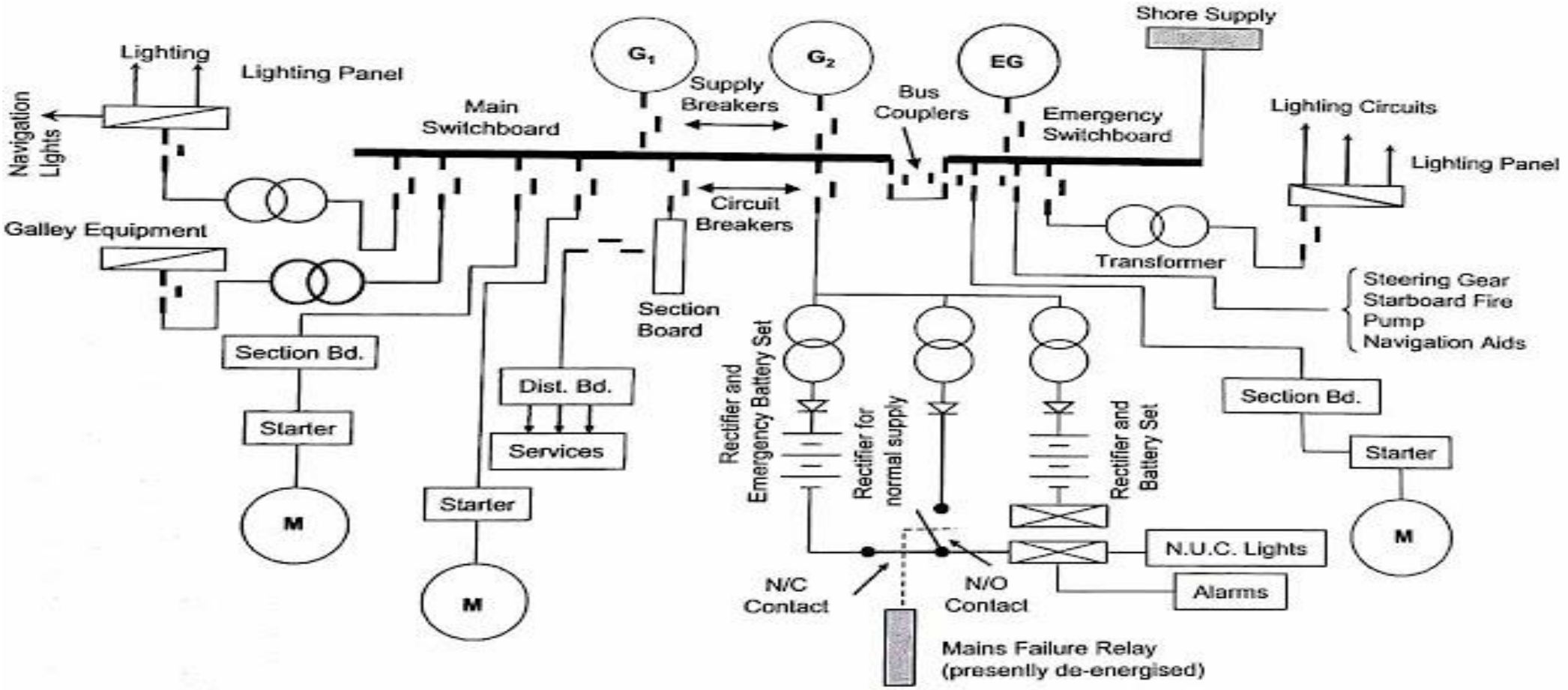
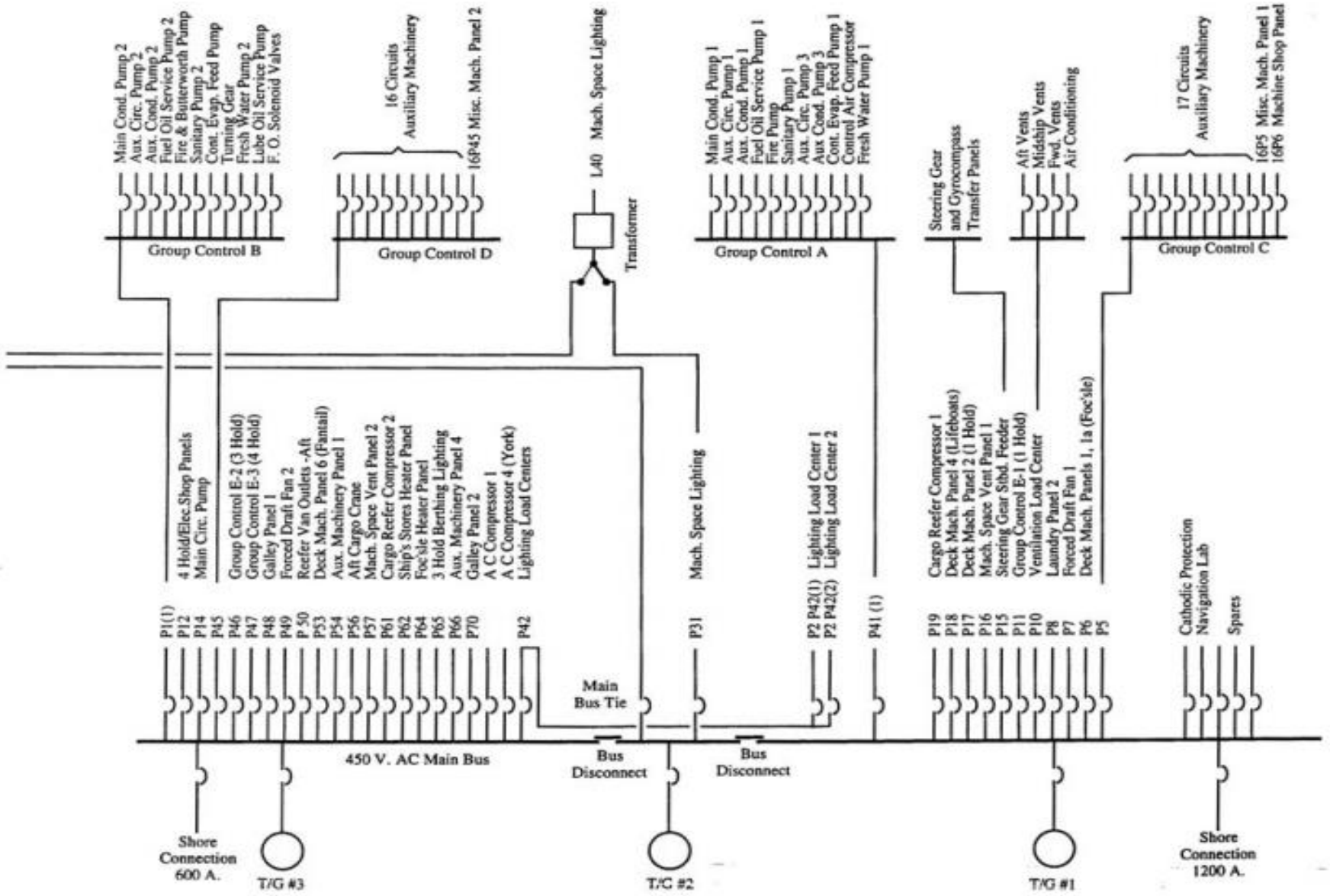
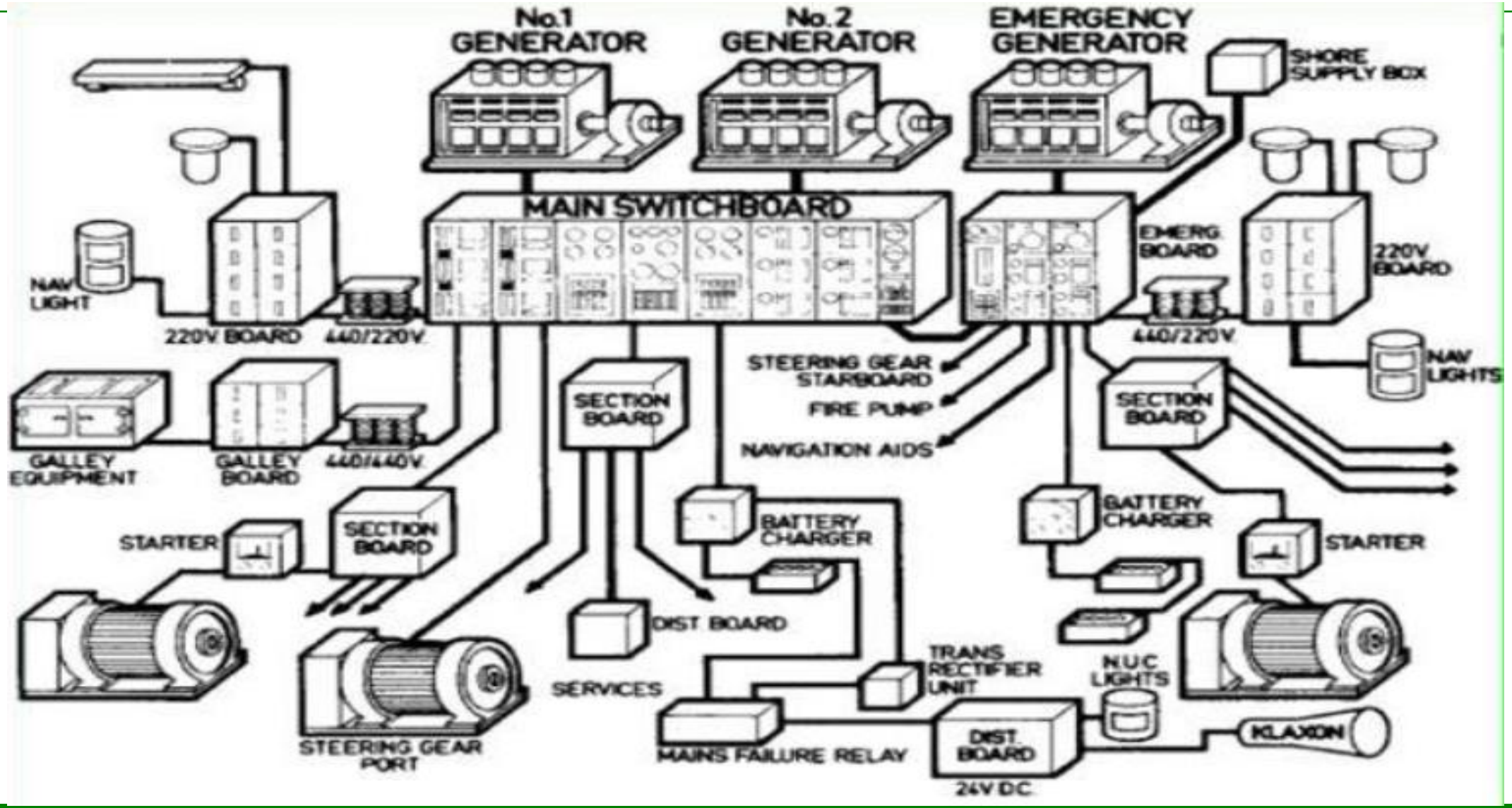


Figure 4.9 – A Ship's Typical Electrical Distribution System



Ship's distribution system

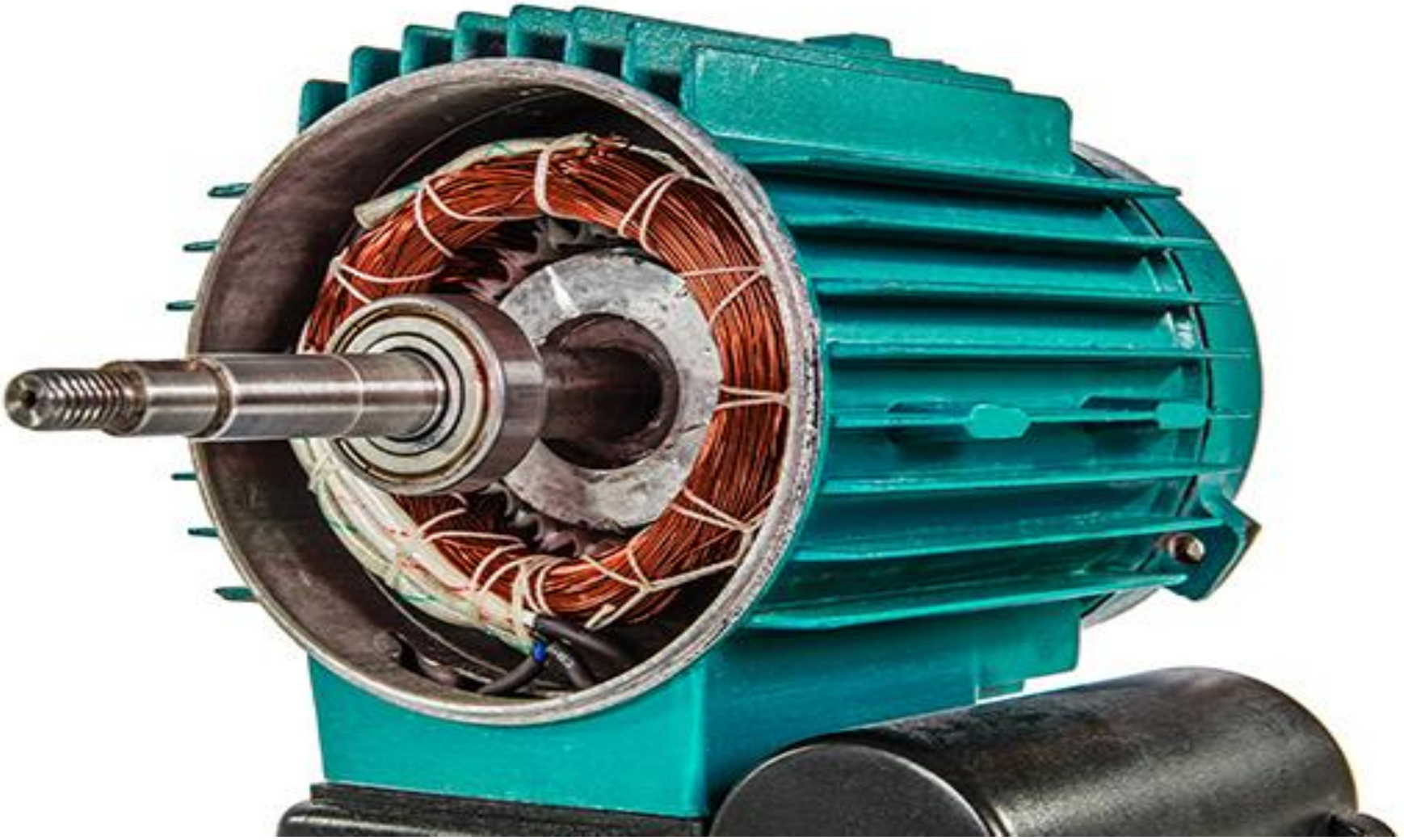

No.1 GENERATOR
No.2 GENERATOR
EMERGENCY GENERATOR
SHORE SUPPLY BOX
MAIN SWITCHBOARD
NAV LIGHT
220V BOARD
440/220V
EMERG BOARD
220V BOARD
NAV LIGHTS
STEERING GEAR STARBOARD
FIRE PUMP
NAVIGATION AIDS
GALLEY EQUIPMENT
GALLEY BOARD
440/440V
SECTION BOARD
SECTION BOARD
BATTERY CHARGER
BATTERY CHARGER
STARTER
SECTION BOARD
STARTER
DIST BOARD
SERVICES
TRANS RECTIFIER UNIT
NUC LIGHTS
KLAXON
STEERING GEAR PORT
MAINS FAILURE RELAY
DIST BOARD
24V DC

Electrical distribution systems



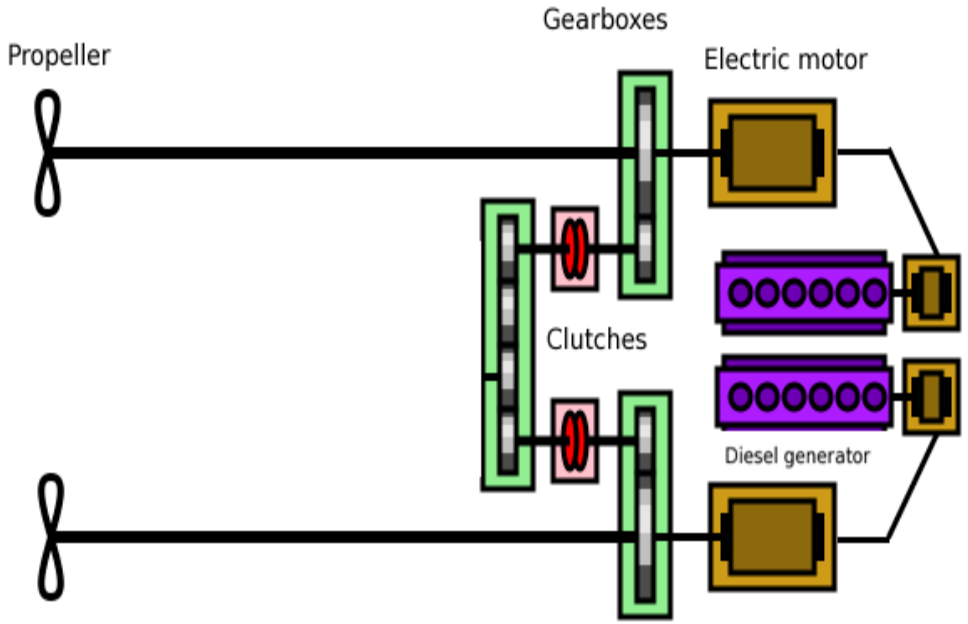
Ship shore supply

Motors



Motors

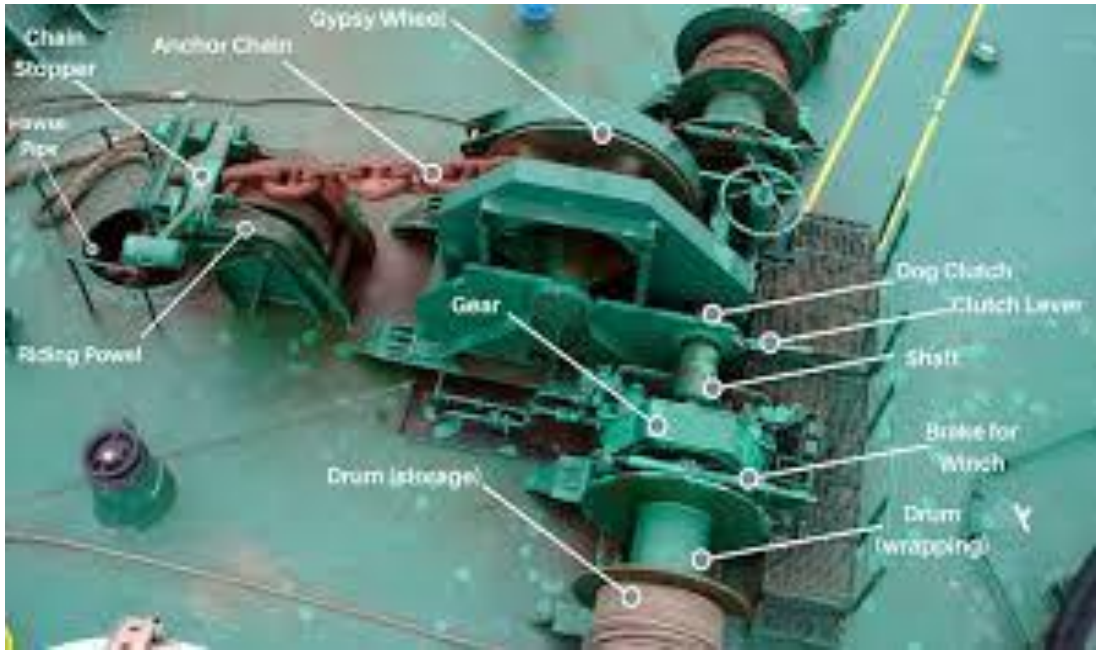
- Uses
  - ✓ Electric propulsion

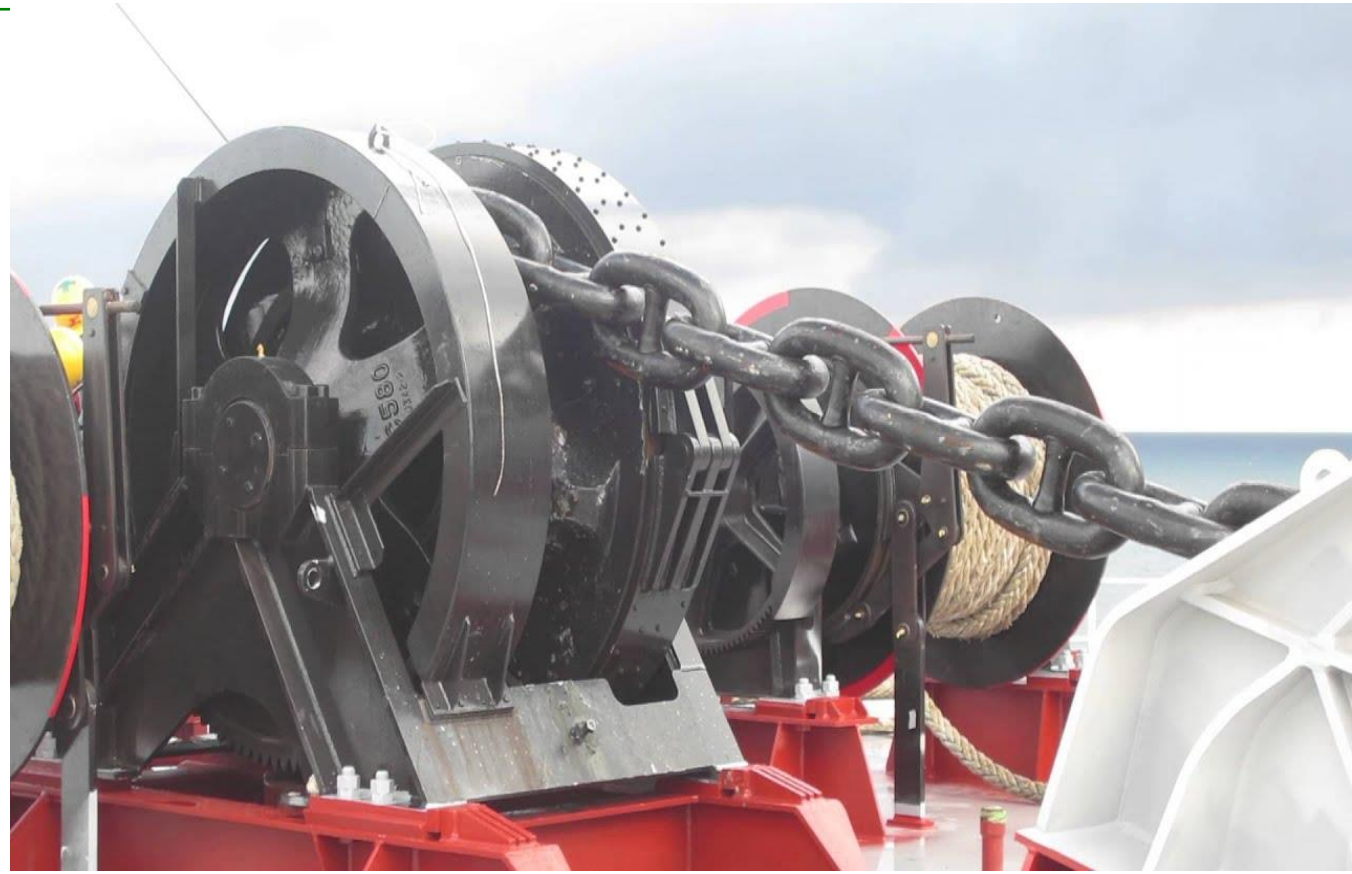




Motors

- ✓ **Windlass**





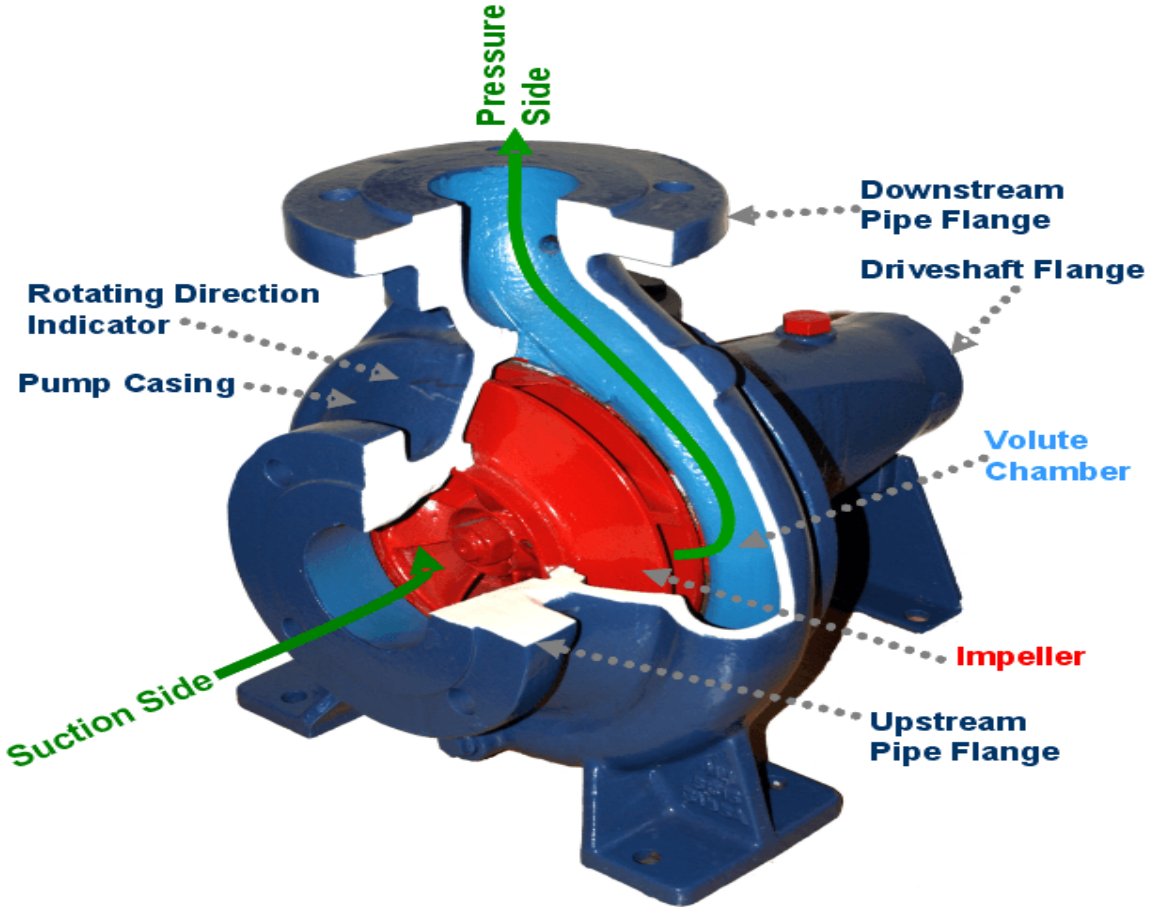
Motors

- ✓ **Winches**



Motors

✓ Pumps





Motors

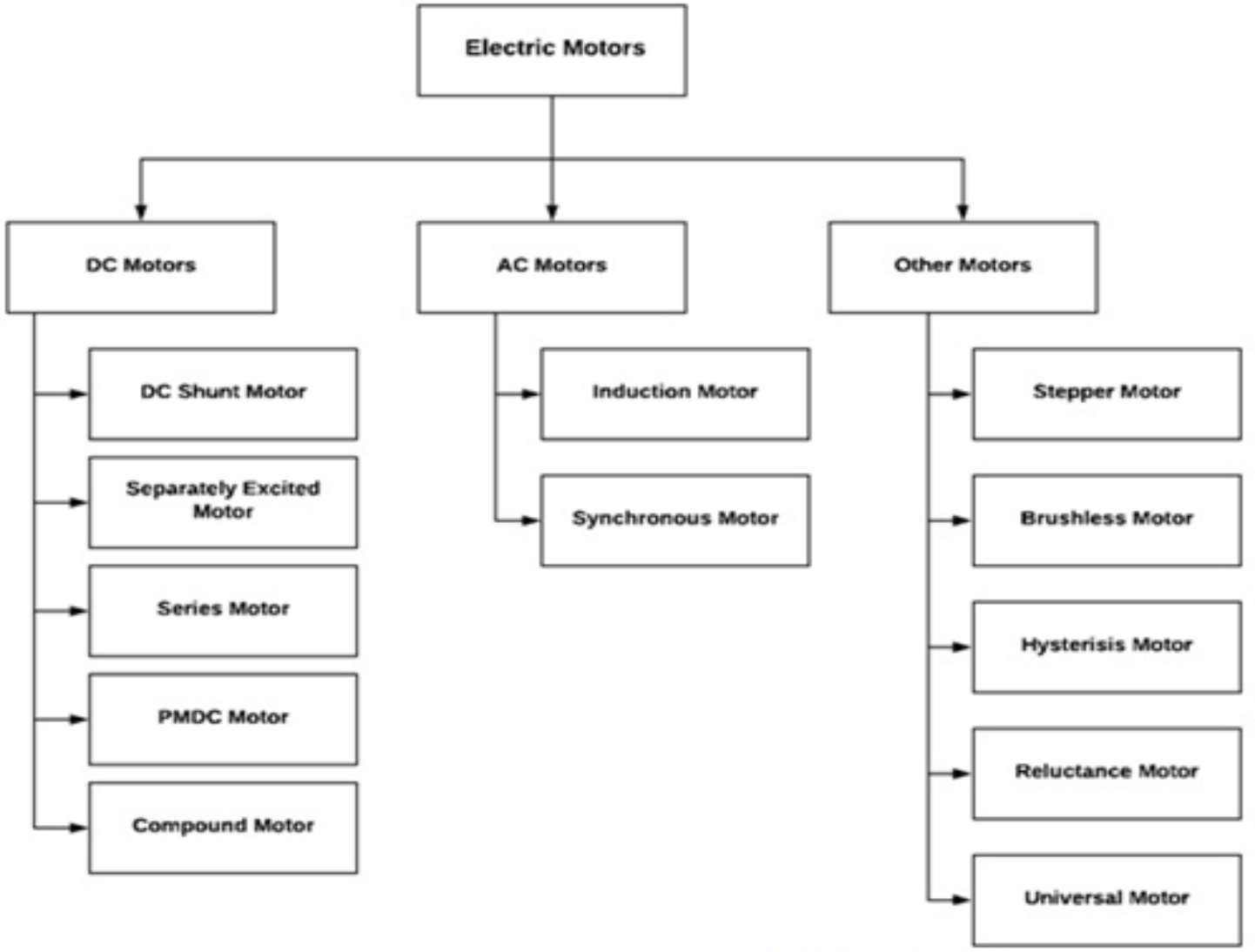
- ✓ **Fans**





Motors

- **Classes of motors**



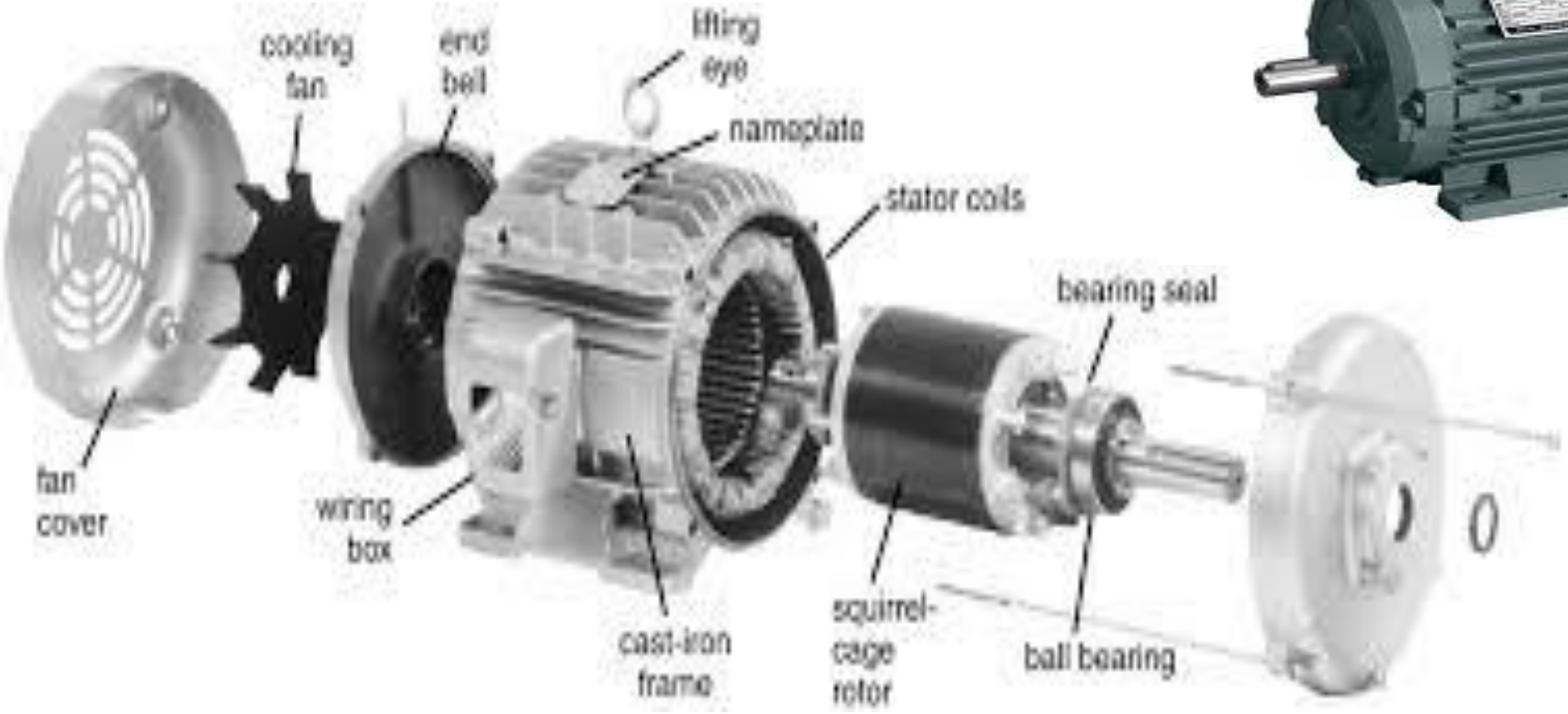
Motors

- **Motors used in marine vessels**
  - ✓**Induction (Asynchronous) Motors**
  - ✓**Synchronous motors**

Induction Motors

- They are the workhorses of marine vessels

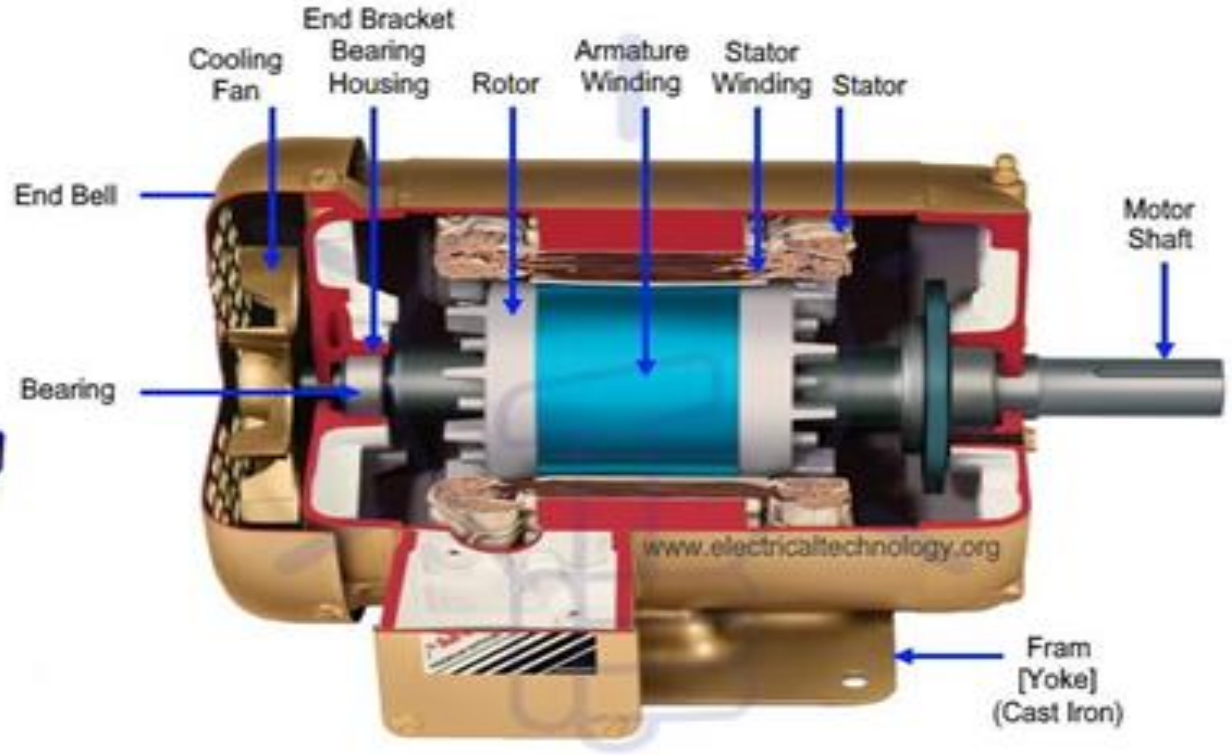




Induction Motors

- They are either single-phase or three-phase type





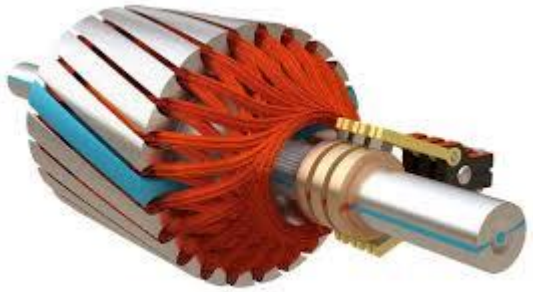
Induction Motors

- **Single-phase induction motors**
  - ✓**They are powered by single-phase power supply.**
  - ✓**They are mostly used in low power applications such as small pumps, small portable compressors and small fans.**

Induction Motors



- Three-phase induction motors
  - ✓They are supplied with 3 three-phase AC supply
  - ✓They are of two types, namely, squirrel cage and wound rotor.
  - ✓Squirrel cage motors are widely used on ships due to their rugged construction and simple design.
  - ✓Typical applications of three-phase induction motors are: lifts, cranes, large capacity exhaust fans, engine auxiliary pumps, engine blower fan motor, engine room heavy load pumps including pumps for fire-fighting, freshwater and sea water.



Induction Motors

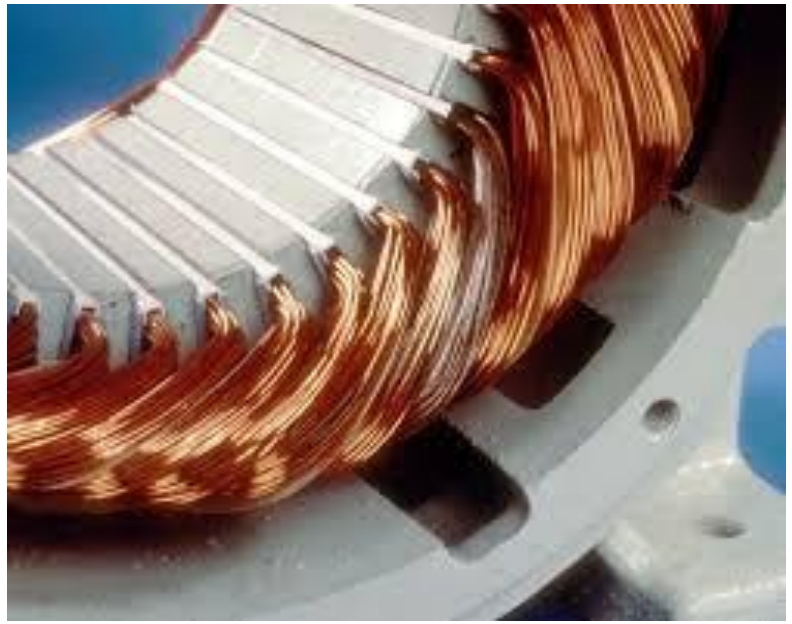
- **Advantages of induction motors**
  - ✓**They are robust and simple in construction, with few moving parts.**
  - ✓**They can efficiently operate in a rugged and harsh environment such as in seagoing vessels.**
  - ✓**The maintenance cost of 3-phase induction motor is less; they do not have parts like brushes, commuters or slip rings.**
  - ✓**An induction motor can operate in an intrinsic environment as they do not have brushes which can cause sparks that could ignite fire.**
  - ✓**3-phase induction motors do not need additional starting mechanism or arrangement as they can generate self-starting torque**

Induction Motors

- **Disadvantages of induction motors**
  - ✓**During starting, they draw high starting current, particularly when attached to a heavy load. This causes a dip in voltage during the starting period of the machine.**
  - ✓**Induction motors operate at lagging power factor which results in increased power losses and efficiency reduction, especially at low load.**

Induction Motors

- Potential faults
  - ✓Environmental faults
    - Moisture in air
    - Vibration from heave equipment
  - ✓Winding faults due to insulation failure
  - ✓Unbalance supply due to blown fuses
  - ✓Crawling due to harmonics in supply
  - ✓Imbalance rotor due to defect in the rotor, internal misalignment, bending of the shaft, uneven loading and improper motor and load coupling.



Induction Motors

- ✓Fatigue failure due to improper maintenance, use of poor quality parts, weakening of material caused by repeated load applications.
- ✓Bearing failure due to poor maintenance, due to overloading, wrong installation, contaminated lube oil, operation at excessive temperature and water in lubrication (grease).
- ✓Lubrication problem due to lack of lubrication or contamination of lubricants leading to can lead to increase friction between parts and bearings.

YOU REAP WHAT YOU SOW

❖ **Galatians 6:7-8**

[7]Do not be deceived, God is not mocked; for whatever a man sows, that he will also reap.

[8]For he who sows to his flesh will of the flesh reap corruption, but he who sows to the Spirit will of the Spirit reap everlasting life.

❖ **Mt 7:12**

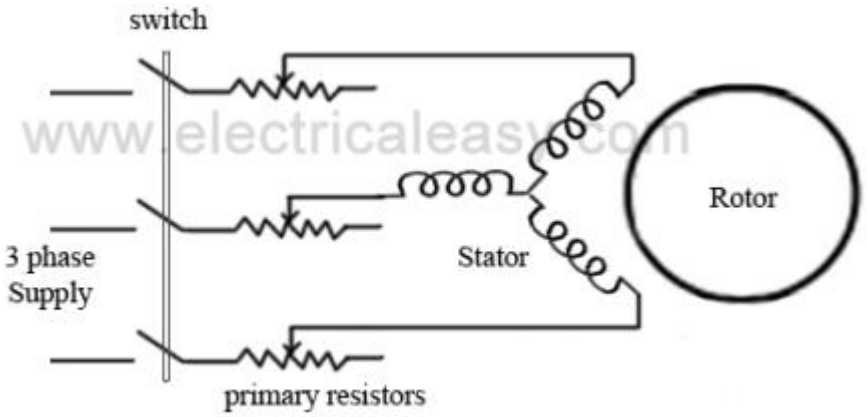
Therefore, whatever you want men to do to you, do also to them, for this is the Law and the Prophets.

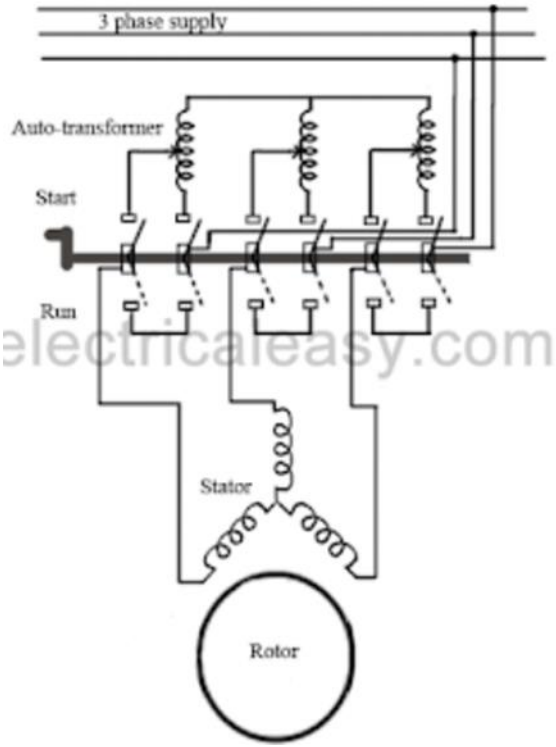
Induction Motors

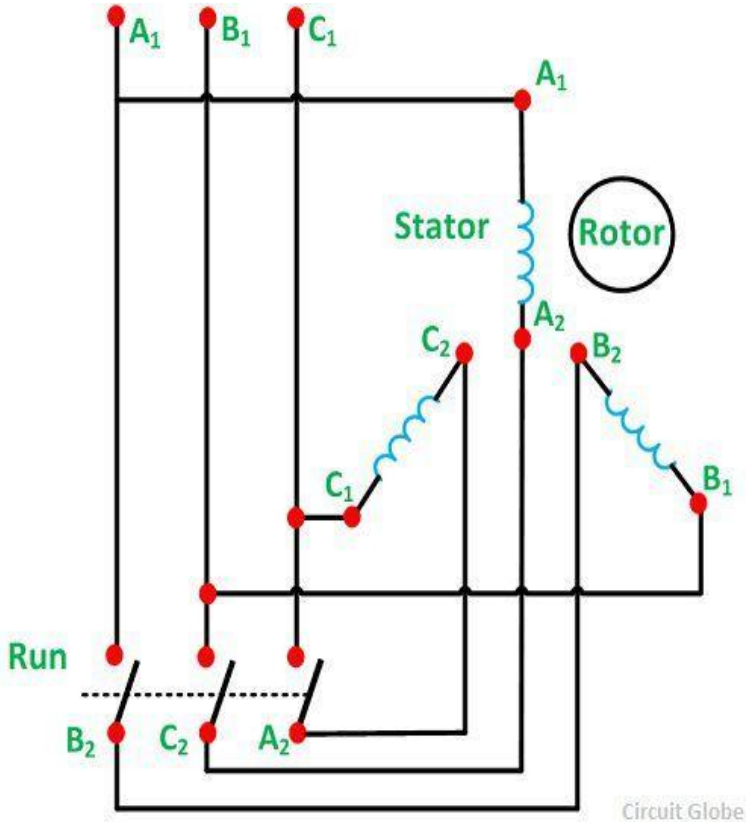
- **Protection of induction motors**
  - ✓**Overload protection**
  - ✓**Overcurrent protection**
  - ✓**Phase unbalance protection**
  - ✓**Stalling (locked rotor) protection**

Induction Motors

- **Starting of induction motors**
  - ✓**Direct-on-line starting**
  - ✓**Resistance starting**
  - ✓**Auto-transformer starting**
  - ✓**Star-Delta starting**

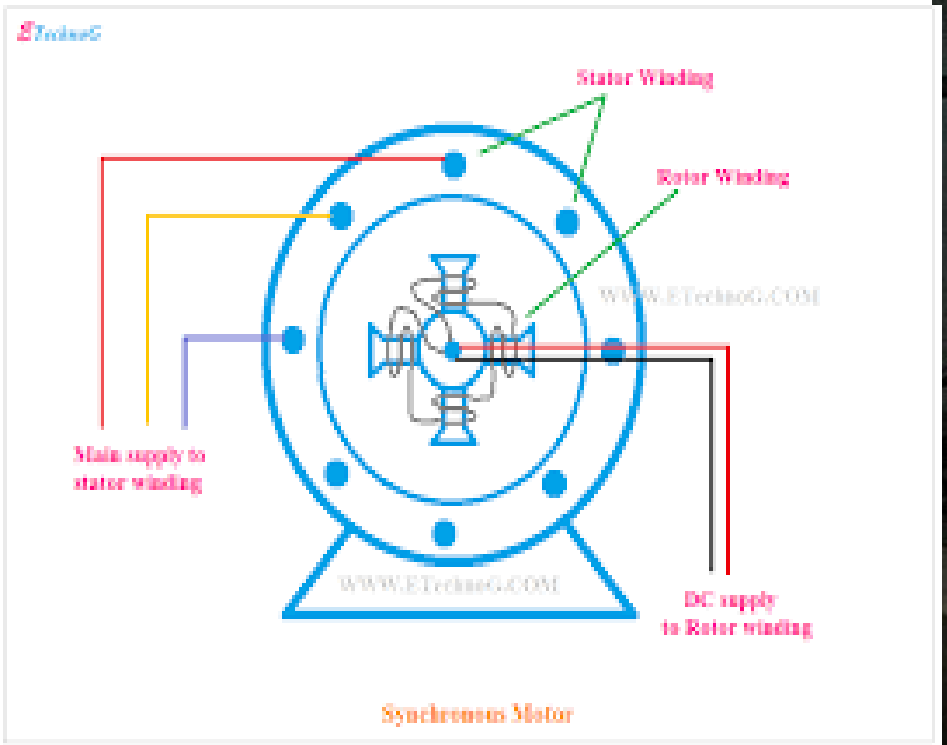


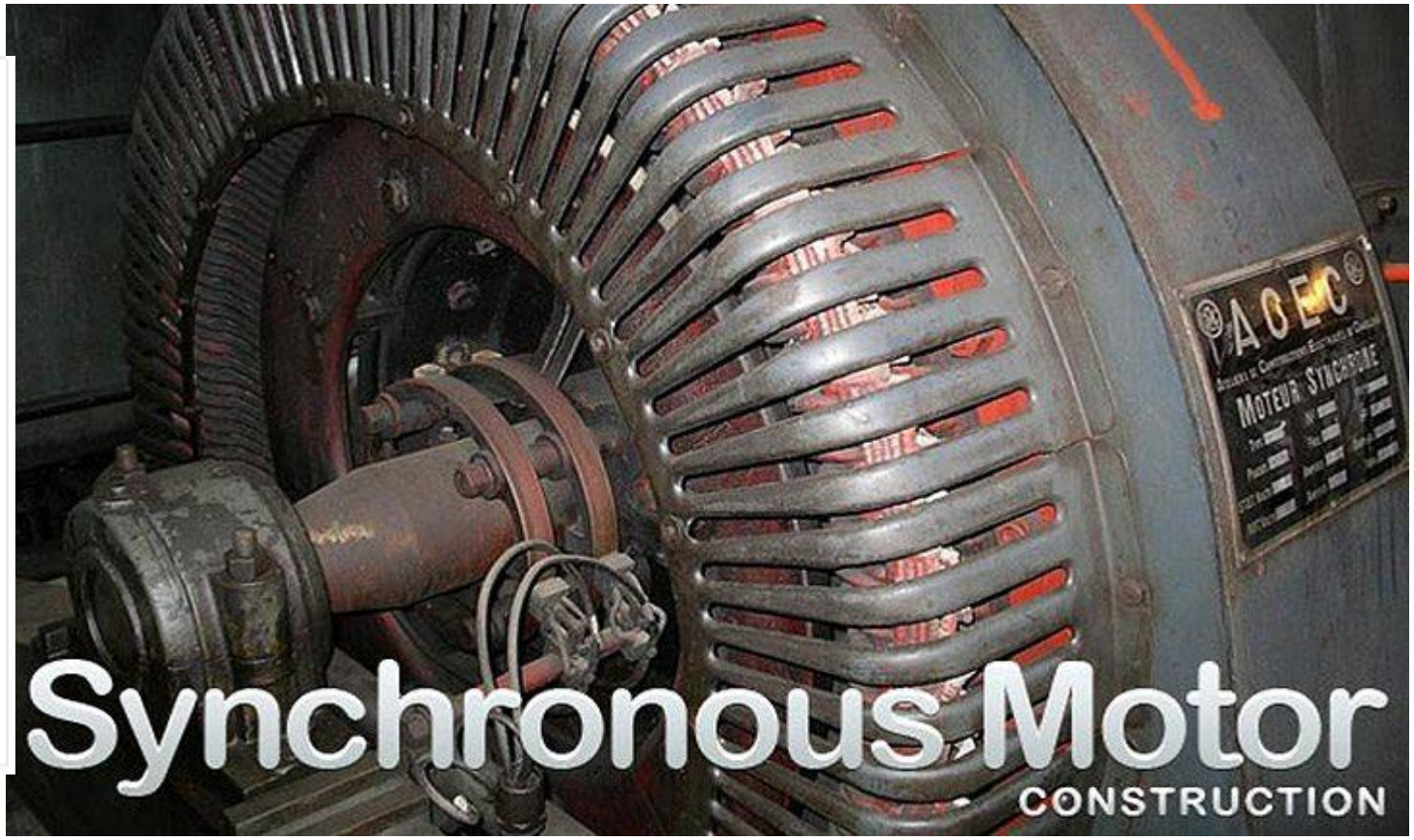




Synchronous Motors

- **They are normally preferred for large power & low speed applications (>5MW).**



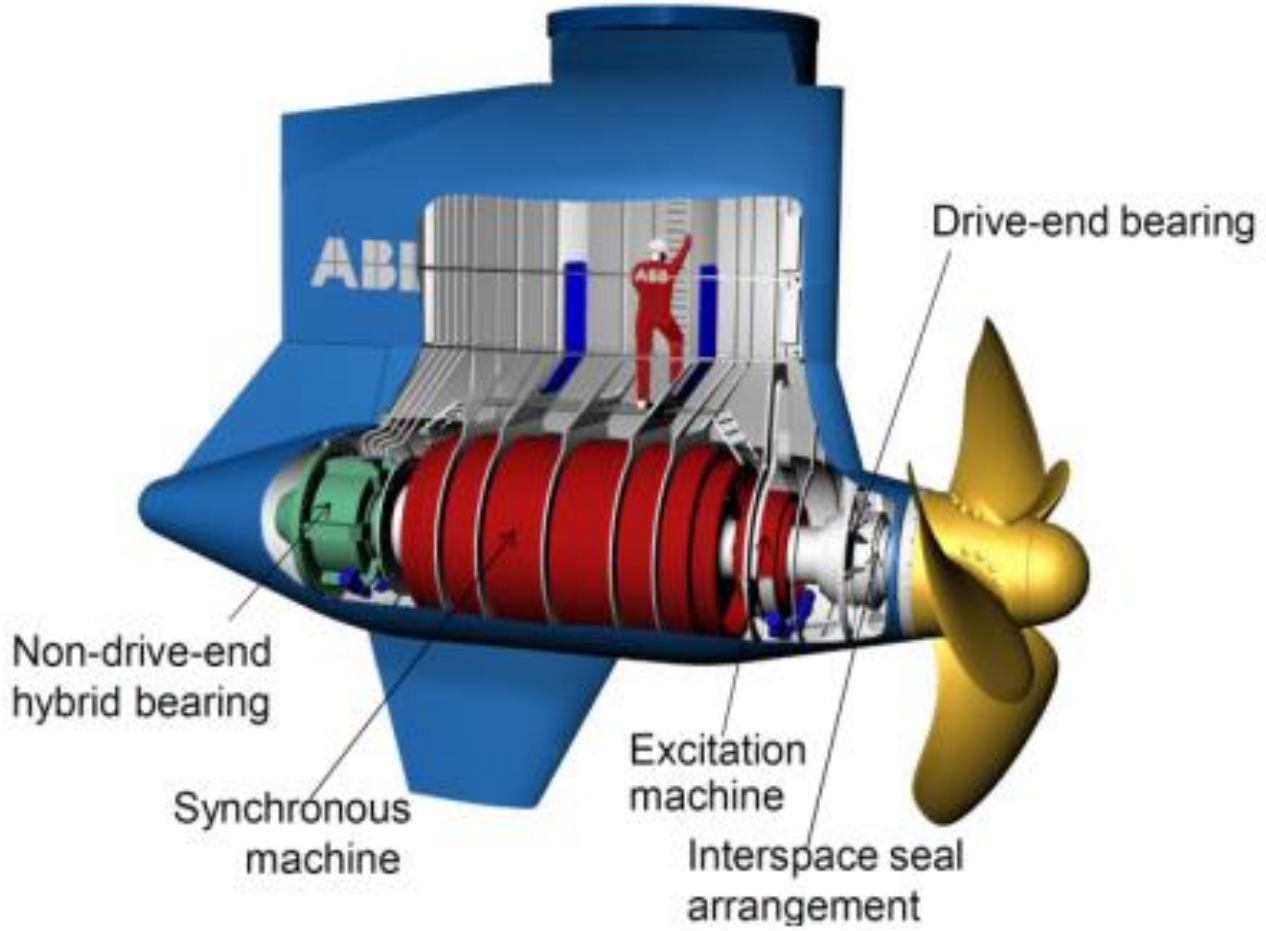


Synchronous Motors



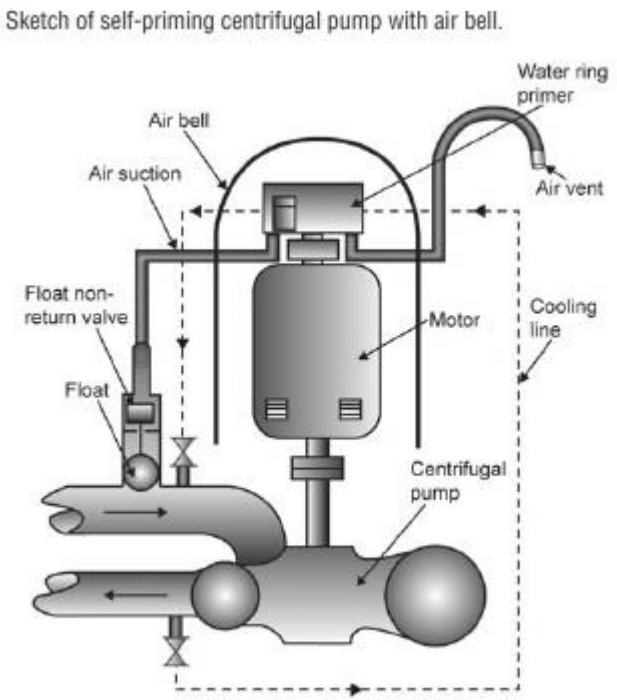
Synchronous Motors

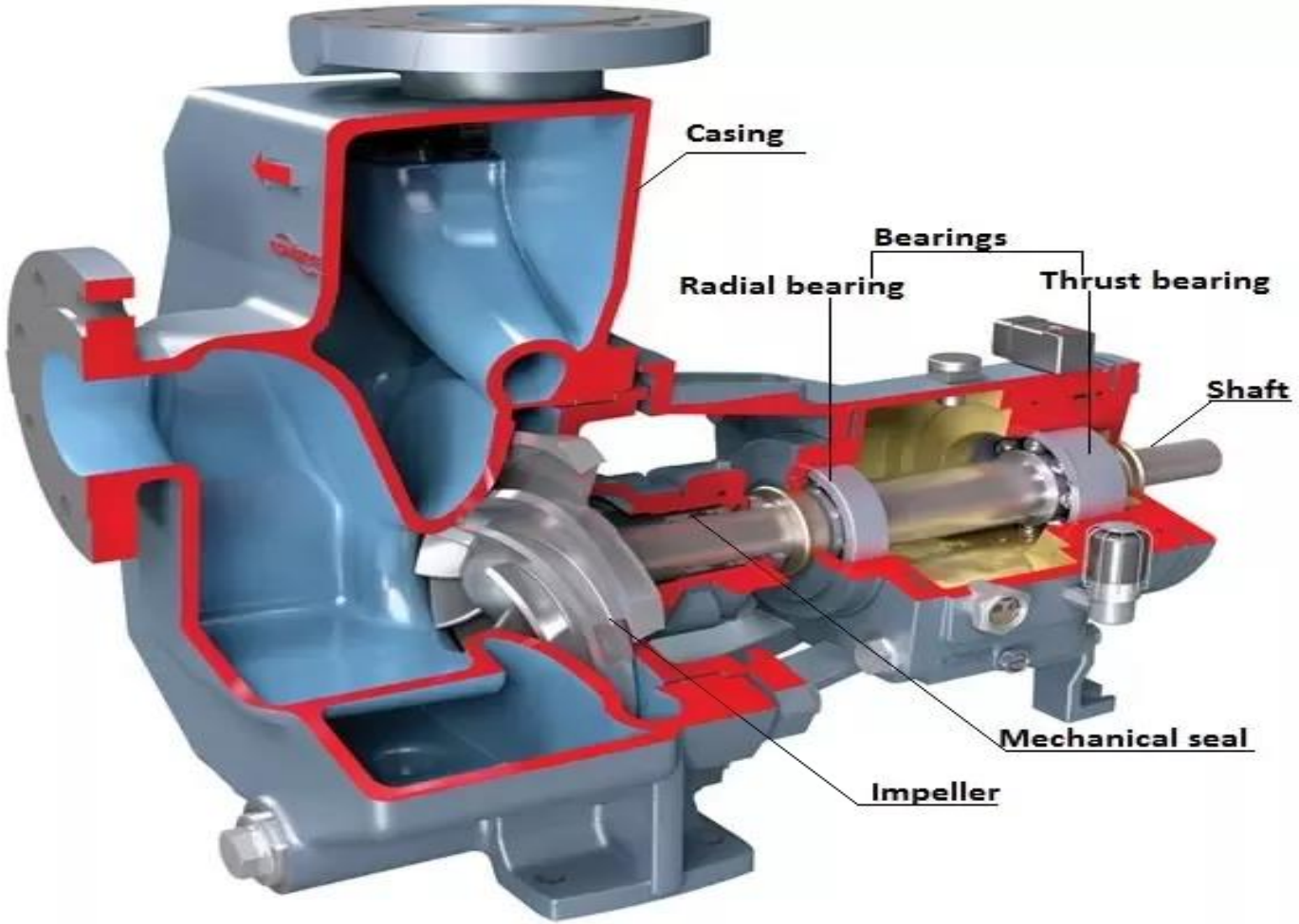
- **Applications in ships**
  - ➢**To drive propeller shaft**



Synchronous Motors

- **Centrifugal pumps**





Synchronous Motors

- **Belt-driven reciprocating compressors**



Synchronous Motors

- **Advantages**
  - **It can control power factor.**
  - **Speed remains constant irrespective of the loads.**
  - **They are built with wider air gaps than induction motors which make them mechanically more stable.**
  - **Synchronous motors usually operate with higher efficiencies (more than 90%) especially in low speed compared to induction motors.**

Synchronous Motors

- **Disadvantages**
  - ➢ **Synchronous motors require dc excitation which is supplied from external sources.**
  - ➢ **They are largely not self-starting motors and need some external arrangement for starting and synchronizing.**
  - ➢ **The cost per kW output is commonly higher than that of induction motors.**
  - ➢ **Unless the incoming supply frequency is adjusted, there is no possible way to adjust the speed.**
  - ➢ **They cannot be started on load because its starting torque is zero.**

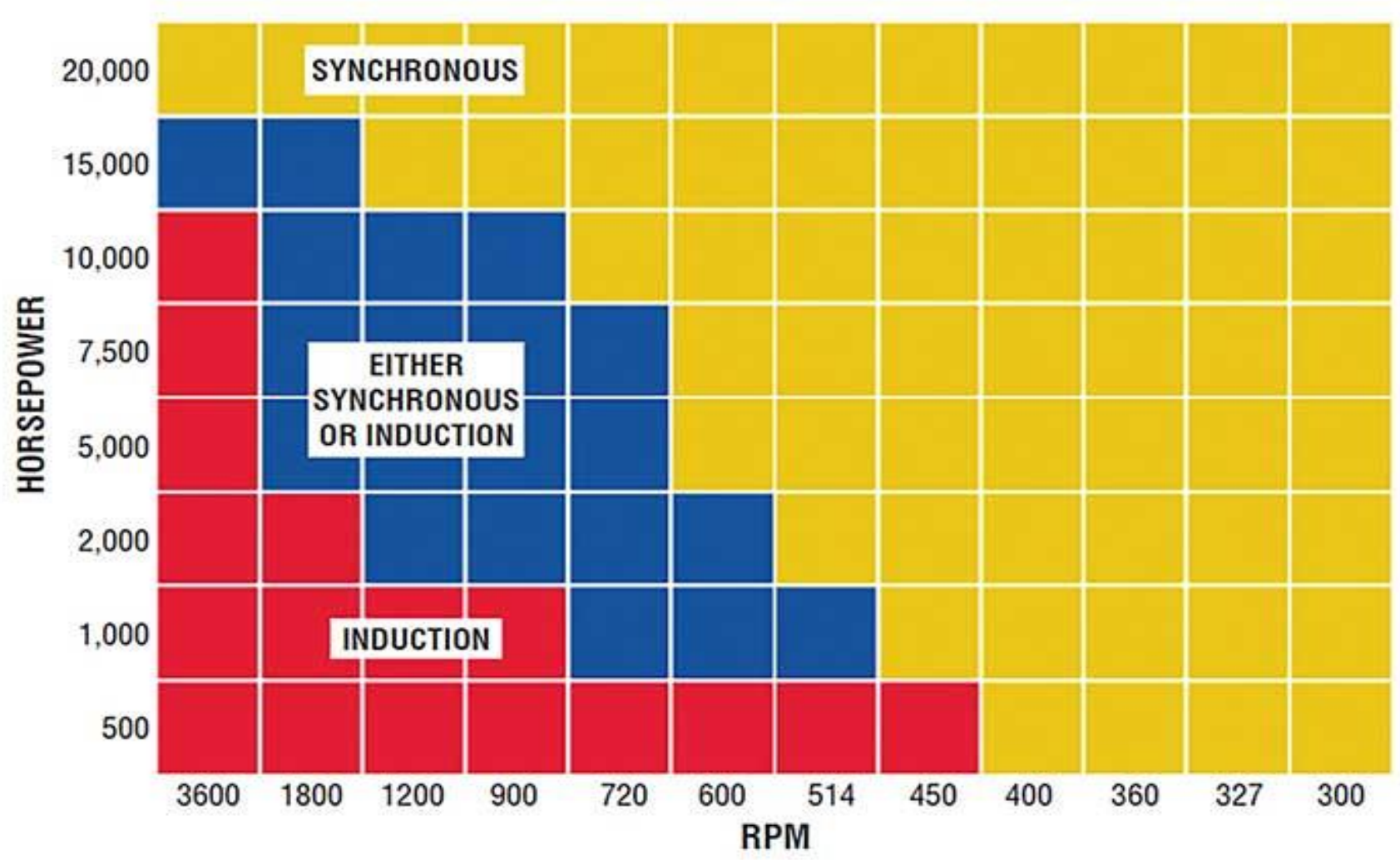
Synchronous Motors

- **Disadvantages**
  - **Collector rings and brushes are required which results in high maintenance cost.**
  - **Synchronous motors cannot be useful for applications requiring frequent starting of machines.**

Synchronous Motors

- **Starting method**
  - **Pony motors**
  - **Damper winding**
  - **Slip ring induction motor**
  - **Small d.c. machine**

Synchronous and Asynchronous Motors Applications

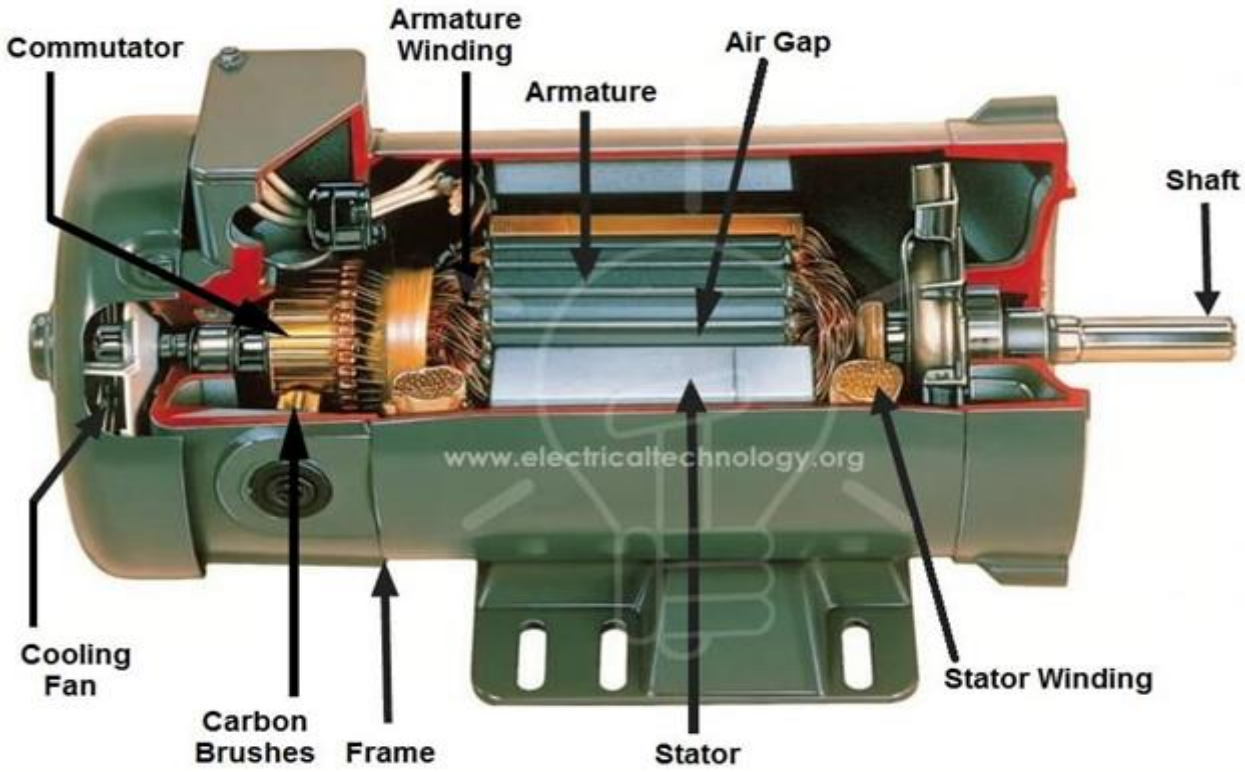


Speed control of Motors

- The direct on-line ship motor will rotate with a speed directly determined by the ship network frequency.
- Variable speed control is necessary for cranes, winches, windlass, capstans, forced-draught fans, etc.
- Types of controllers:
  - AC controller (Variable frequency drive(VFD)
    - ✓Current source inverter (CSI)
    - ✓Voltage source inverter
    - ✓Pulse width modulation
  - Cycloconverter

DC Motors

- The are used when constant speed is required.
- Their applications are in lifts, blowers, fans, conveyors, etc.



Earthing

- It is a deliberate connection between an electrical power grid and earth.
- It facilitates the detection and discriminative isolation of faults involving contact to earth, and to limit over-voltages under transient conditions.
- On board a metallic ship, earth is the hull and superstructure.
- Earthing of power systems and equipment is done at the neutral.
- There are five methods of neutral earthing.
  - Unearthed neutral system
  - Solid neutral earthed system
  - Resistance neutral earthing system
  - Resonant neutral earthing
  - Transformer earthing

Earthing

- Onboard a ship, the priority is to maintain continuity of the electrical supply to essential equipment in event of a single earth fault.
- The priority requirement ashore is the immediate isolation of earth-faulted equipment.
- Shipboard main LV systems at 440V are normally provided with neutral insulated system.
- HV systems (1000V to 3.3kV) are usually provided with both insulated neutral and neutral earthing via a resistor.
- The international convention for the Safety of Life at Sea (SOLAS) requires that the hazardous areas of tanker, viz: cargo area, pump room should have a neutral insulated system to prevent any stray earth current from flowing in the hull that could cause explosions.

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Earthing

- **Advantages of insulated neutral system**
  - ➢ **Continuity of service on occurrence of an earth fault.**
  - ➢ **Earth fault currents can be kept low.**

- **Disadvantages of insulated neutral system**
  - ➢ **Occurrence of high transient over – voltages.**
  - ➢ **Requirement of high level of insulation.**
  - ➢ **Difficulty in detecting earth faults.**

Earthing

- **Advantages of earthed neutral system**
  - ➢ **Equipment insulation does not require any special attention.**
  - ➢ **Earth faults are detected automatically, and immediately isolated.**
  - ➢ **Earth fault current flows for a short period of time thereby restricting damage.**
  - ➢ **Arcing earth over-voltages can be avoided.**

- **Disadvantages of insulated neutral system**
  - ➢ **Immediate disconnection and loss of service when an earth fault occurs.**
  - ➢ **High earth fault currents flow, which can cause extensive damage and explosion.**

Earthing

- **Causes of earth faults**
  - **dampness,**
  - **mechanical damage,**
  - **contamination due to dirt,**
  - **temperature rise, and**
  - **ageing.**

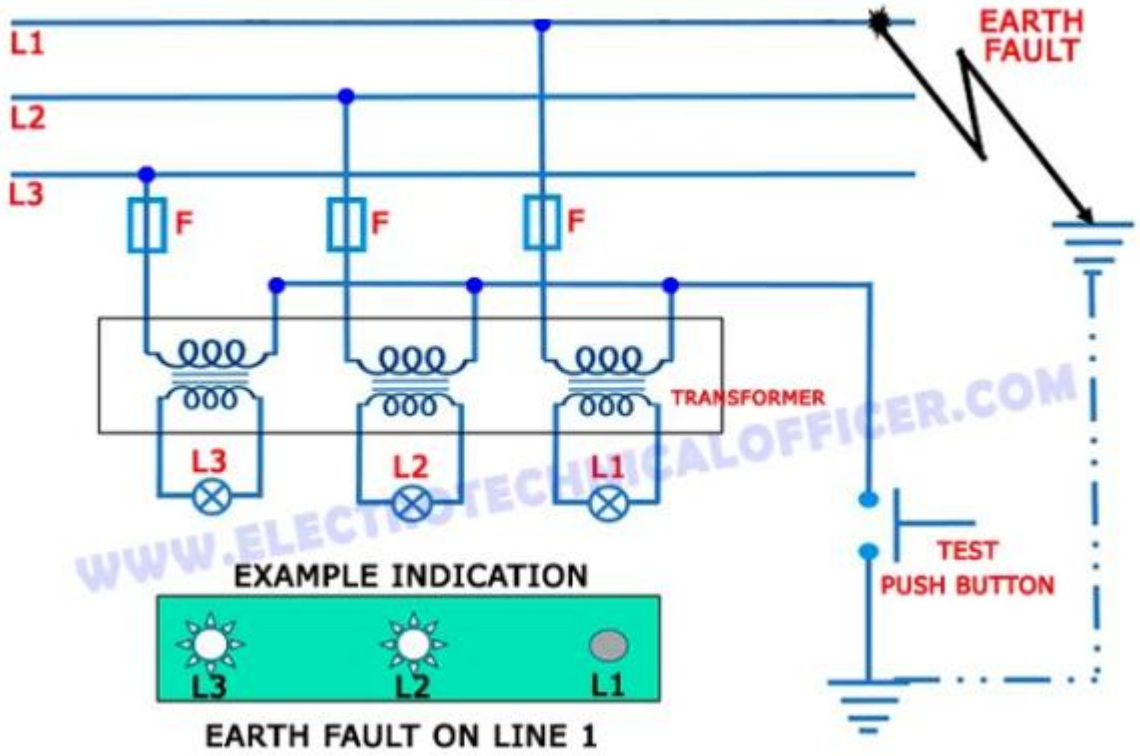
- **Earth faults can be prevented by:**
  - **using appropriate enclosures to prevent the ingress of moisture and dust,**
  - **offering protection against mechanical damage,**

Earthing

- **Earth faults can be prevented by:**
  - **using exact cable glands and seals on cable ends,**
  - **regularly performing insulation resistance tests, and**
  - **strictly maintaining equipment.**

Earthing

- **Detection of Earth faults can be prevented by:**
  - **Regulations require that earth fault indicators are fitted to main switch boards to indicate the presence of an earth fault on each isolated section of a distribution system.**

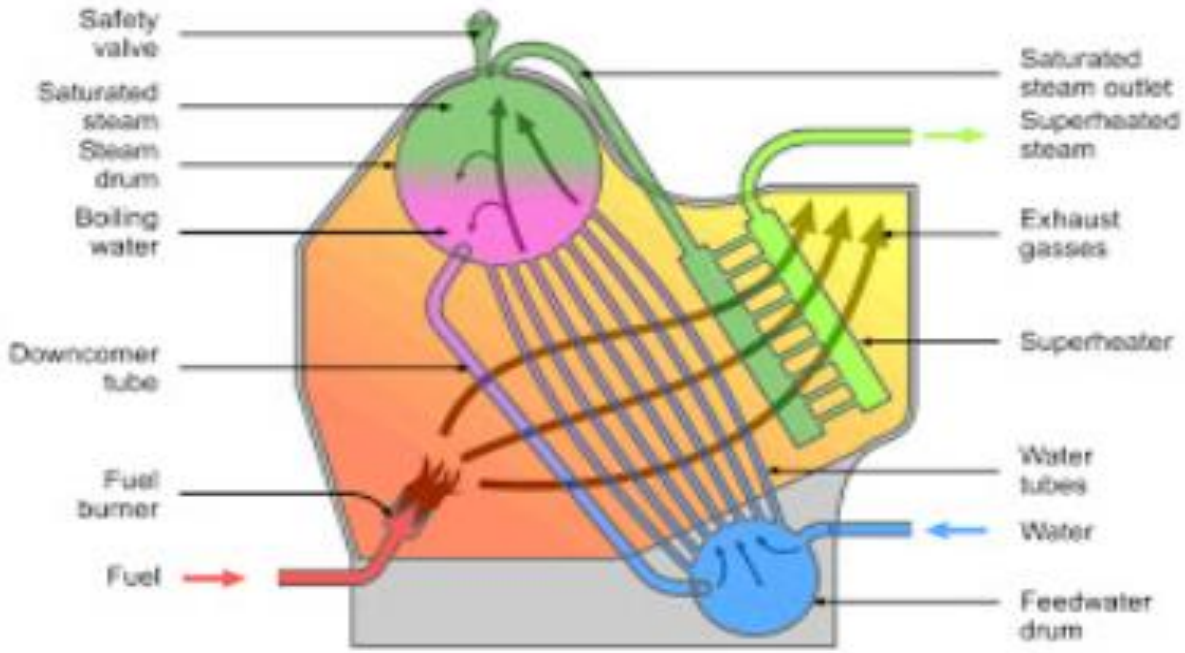


Earthing

- **Common locations of earth faults**
  - **Lamp fittings on the open deck.**
  - **At the laundry, washing machines drenched with water.**
  - **At the galley, the ovens, hot plates, etc.**
  - **Electrical machinery with water dripping on them.**
  - **Overheated motors leading to the melting of varnish on the windings.**
  - **Dirty electrical apparatus with surface tracking (leakage current)**
  - **Terminal lock nuts becoming loose due to vibration etc., leading to cable touching the body.**

Boilers

- They are provide heating to main diesel propulsion engines
- Steam generated by boiler are used for cleaning, and heating of seawater in freshwater generators thus evaporating it to make it potable water.
- The two types of boilers used in marine vessels are water-tube boilers and fire-tube boilers



Boilers

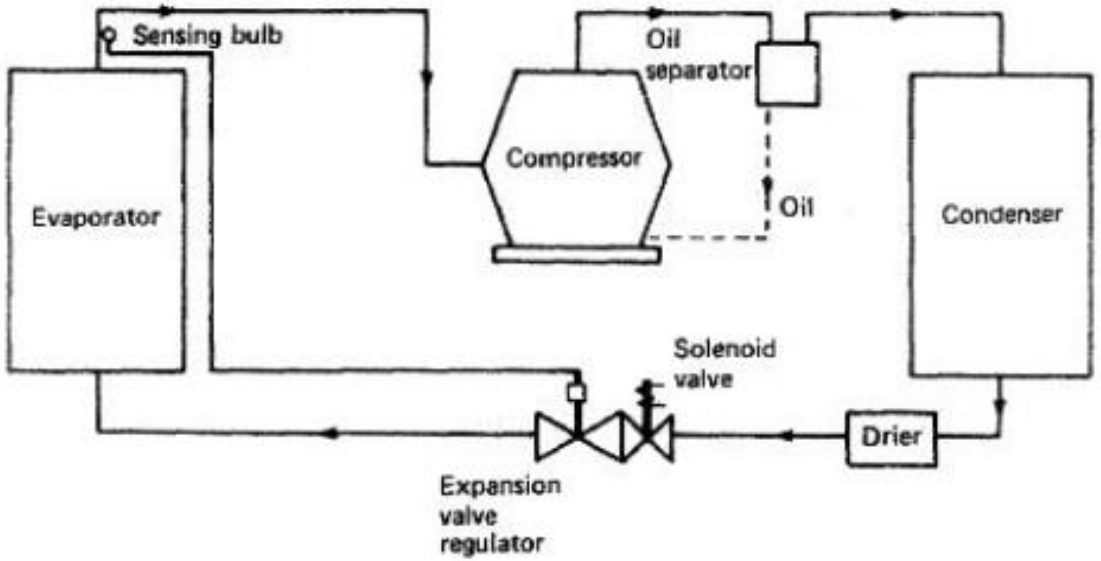
- When a boiler is started, its lighten-up rate must not be accelerated too much as this might cause an unnecessary strain of the boiler material by quick and uneven temperature rises.
- It might be necessary to perform several start/stop sequences to reduce the lighten-up rate.
- At a commissioning start-up of a boiler and after any repair work of a refractory, it is important to further reduce the lighting-up rate owing to the small amount of residual water.
- When heated, the water vaporizes and expands which might cause cracks in the refractory.

Refrigeration plant

- Refrigeration is a process by which the temperature of a space or its contents is reduced to levels below that of their surroundings.
- Refrigeration plants on merchant vessels play a vital part in carrying refrigerated cargo and provisions for the crew on board.
- Refrigeration prevents growth of micro-organisms, oxidation, fermentation and drying out of cargo, etc.
- Refrigeration is also required to cool bulk CO2 for fire-fighting systems.
- Ships refrigeration plant may vary from small domestic refrigerating unit, for provisions, to large plants, for reefer vessels.

Refrigeration plant

- There are four basic components of a refrigeration system.
  - Compressor
  - Condenser
  - Expansion valve
  - Evaporator
- The usual system employed for marine refrigeration plants is the vapour compression cycle



Safety devices on refrigeration plant

- **Low pressure (LP) cut out**
  - ➢ **It cuts off the compressor in the event of pressure drop in the suction line.**
  - ➢ **A drop in pressure below a set value is an indication of optimal cooling.**
  - ➢ **When the pressure rises, indicating there is flow of refrigerant in the line due to increase in room temperature, the LP switch will start the compressor.**

Safety devices on refrigeration plant

- **High pressure or HP cut out**
  - ➢ **It activates and trips the compressor when the discharge side pressure increases above a set limit.**

- **Oil differential cut out**
  - ➢ **In the event of low supply or no supply of lube oil to the bearing, the differential pressure will increase and activates a trip signal to safeguard bearing and crankshaft.**

- **Relief valve**
  - ➢ **It is fitted on the discharge sides of compressors. They safeguard compressors in the event of over pressure.**

Safety devices on refrigeration plant

- **Solenoid valve**
  - ➢ **In case of sudden stoppage of compressor, the master solenoid closes, avoiding the flooding of evaporator with refrigerant.**

- **Oil heater**
  - ➢ **Oil heater is provided for compressor crank case oil and prevents compressor from getting excessively cold, which may affect the lubrication of parts.**